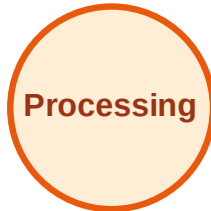


BFS Traversal

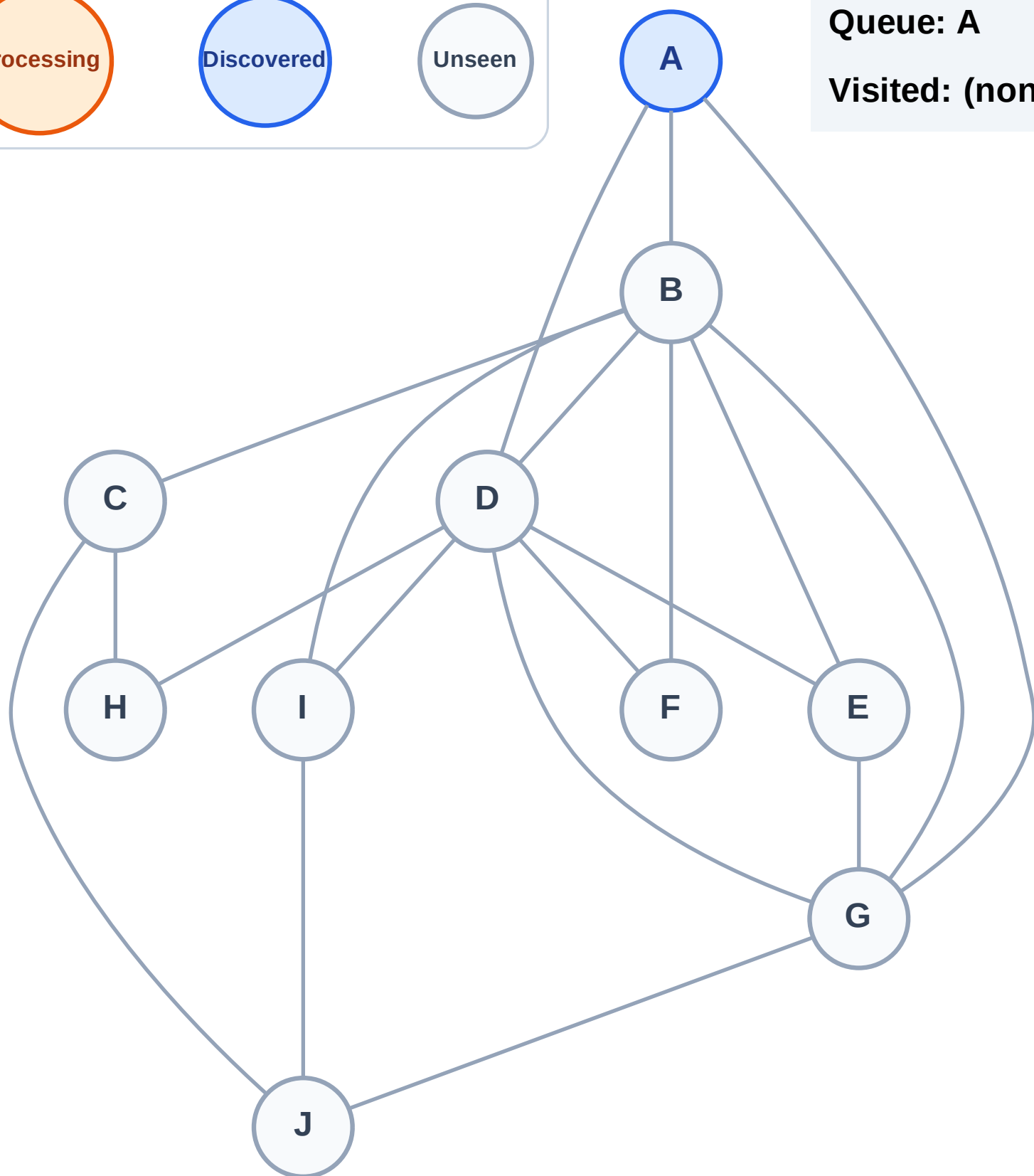
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

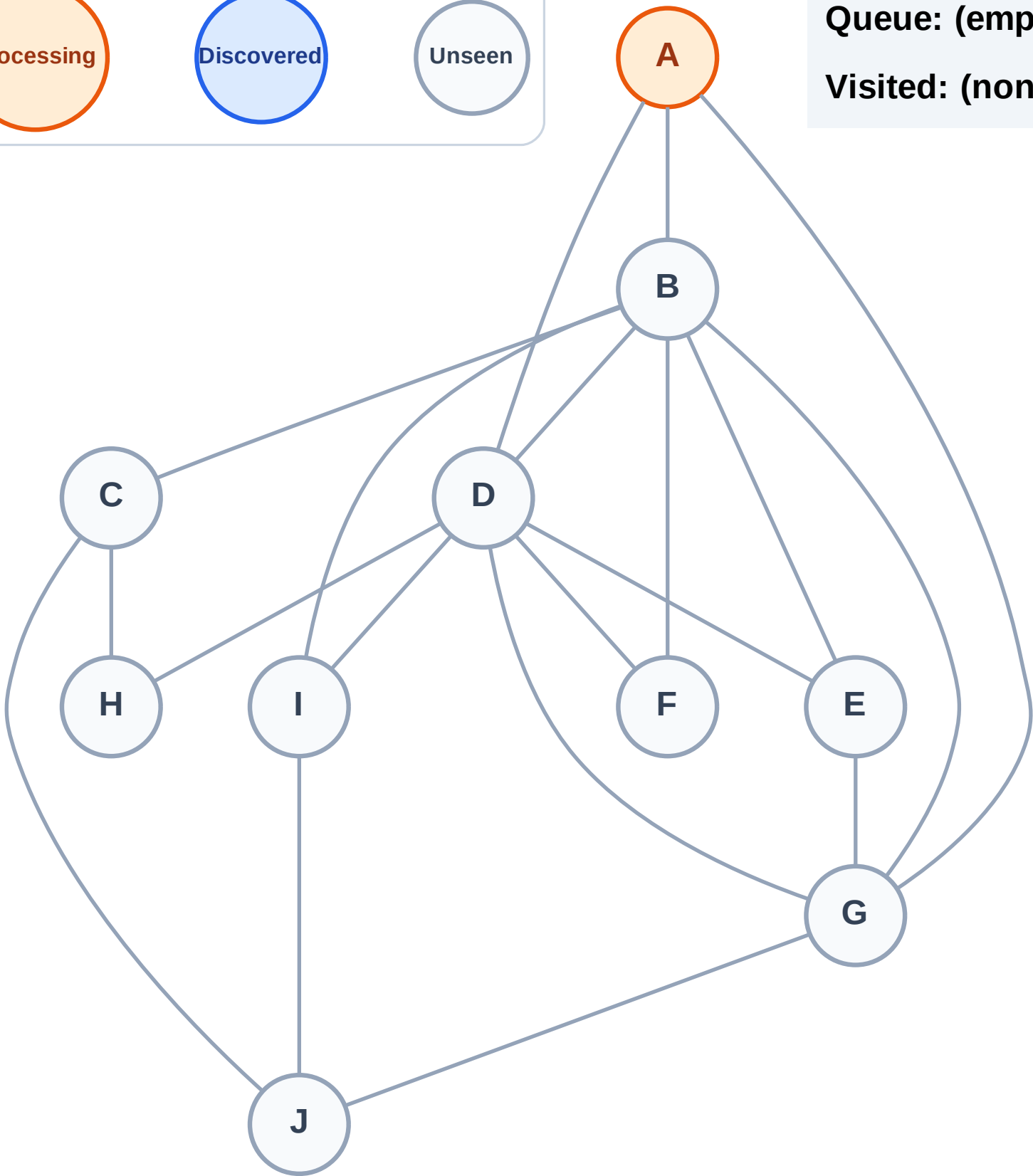
Step 2: Process node A

Legend



Queue: (empty)

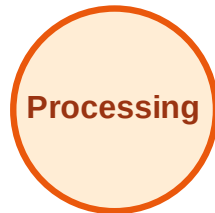
Visited: (none)



BFS Traversal

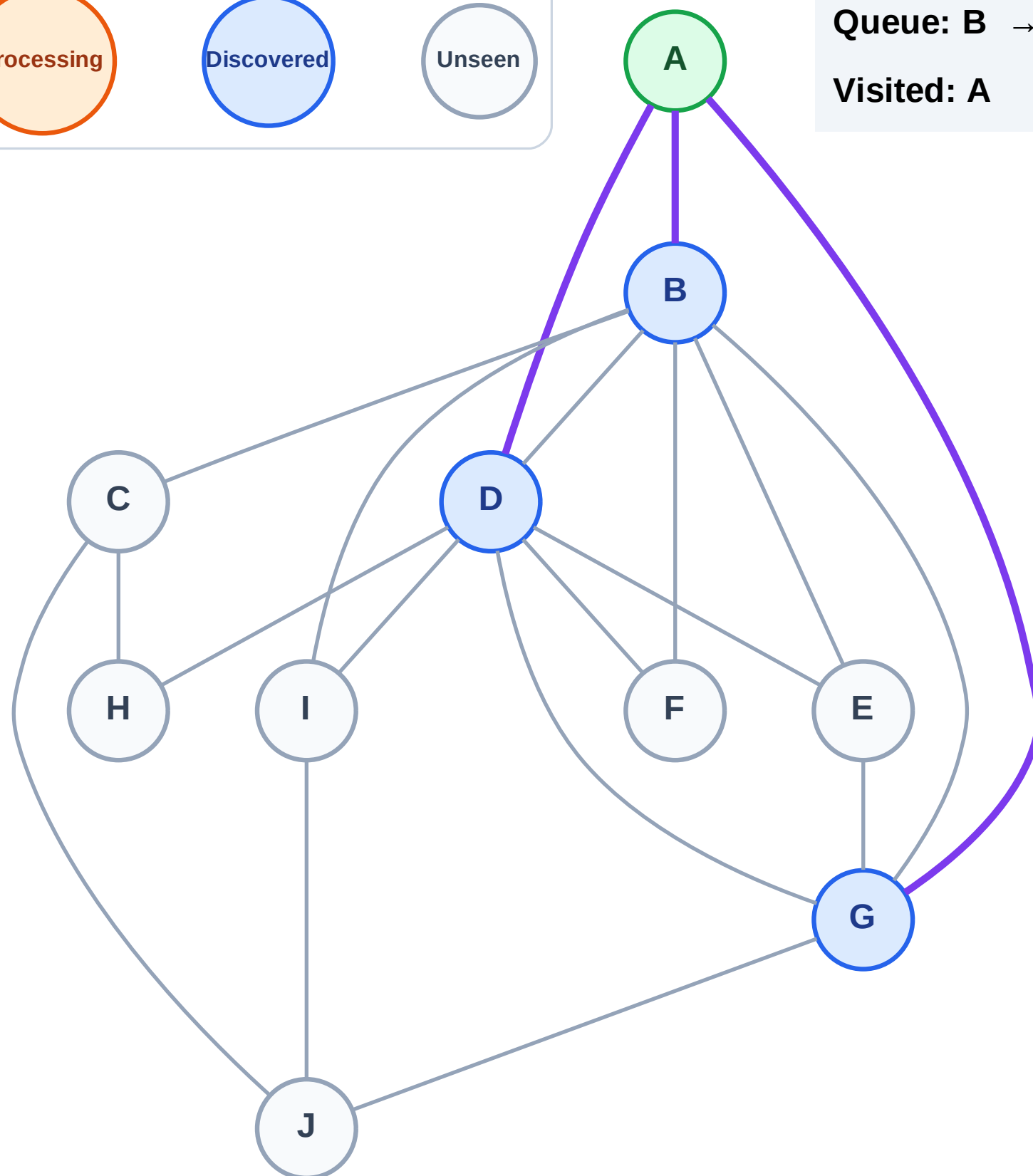
Step 3: Discover B, D, G from A

Legend



Queue: B → D → G

Visited: A



BFS Traversal

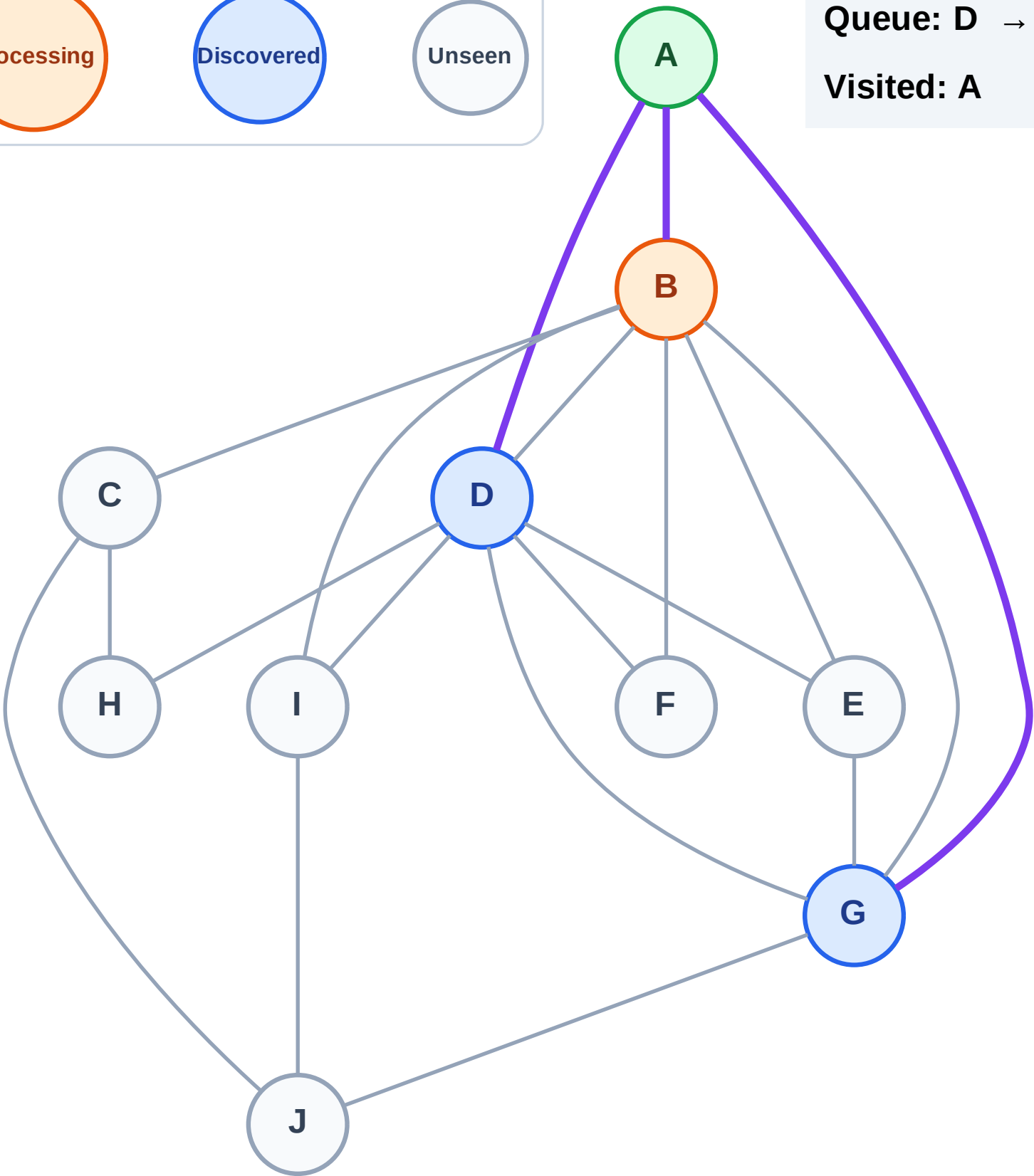
Step 4: Process node B

Legend



Queue: D → G

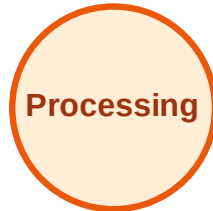
Visited: A



BFS Traversal

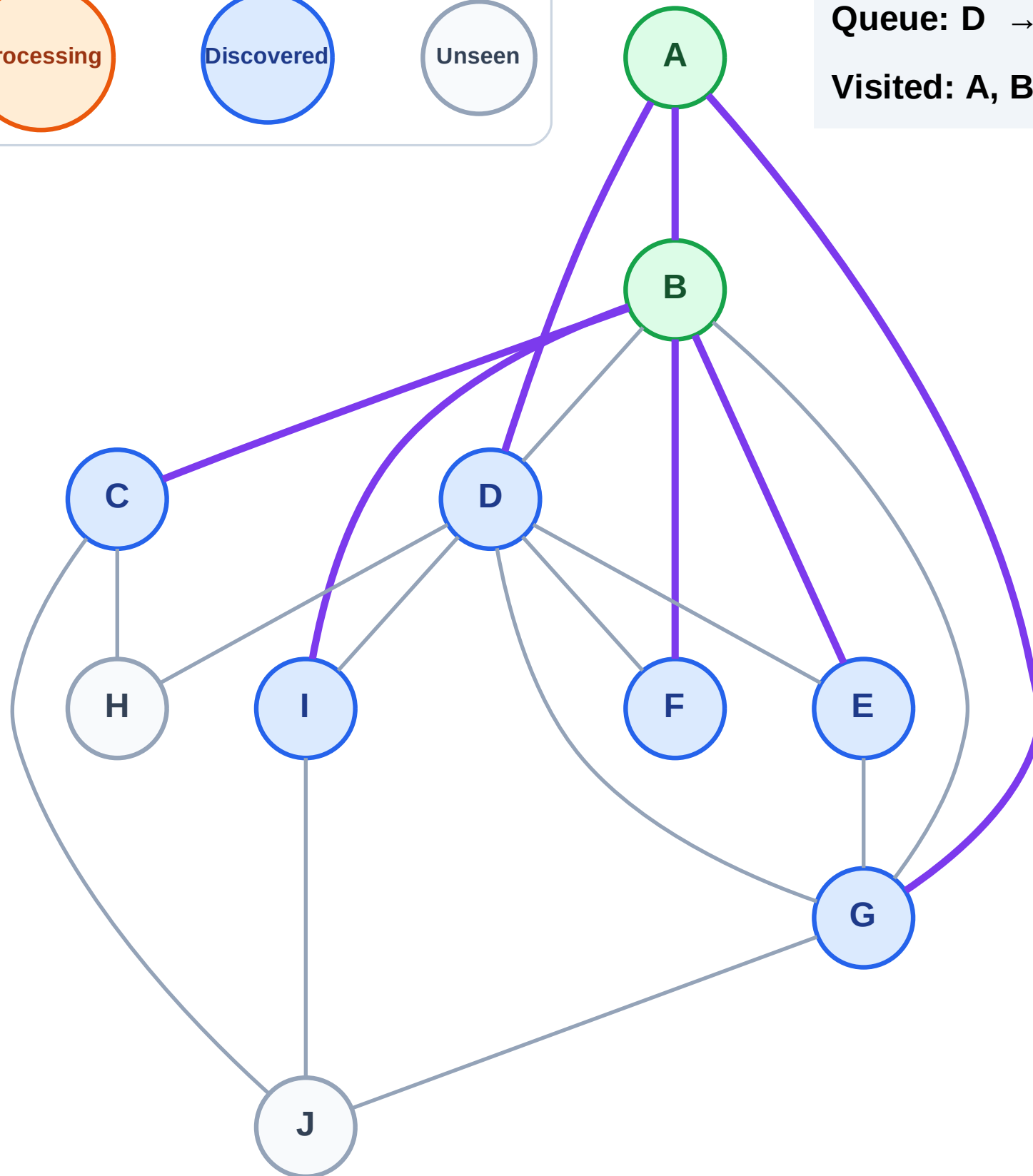
Step 5: Discover C, E, F, I from B

Legend



Queue: D → G → C → E → F → I

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

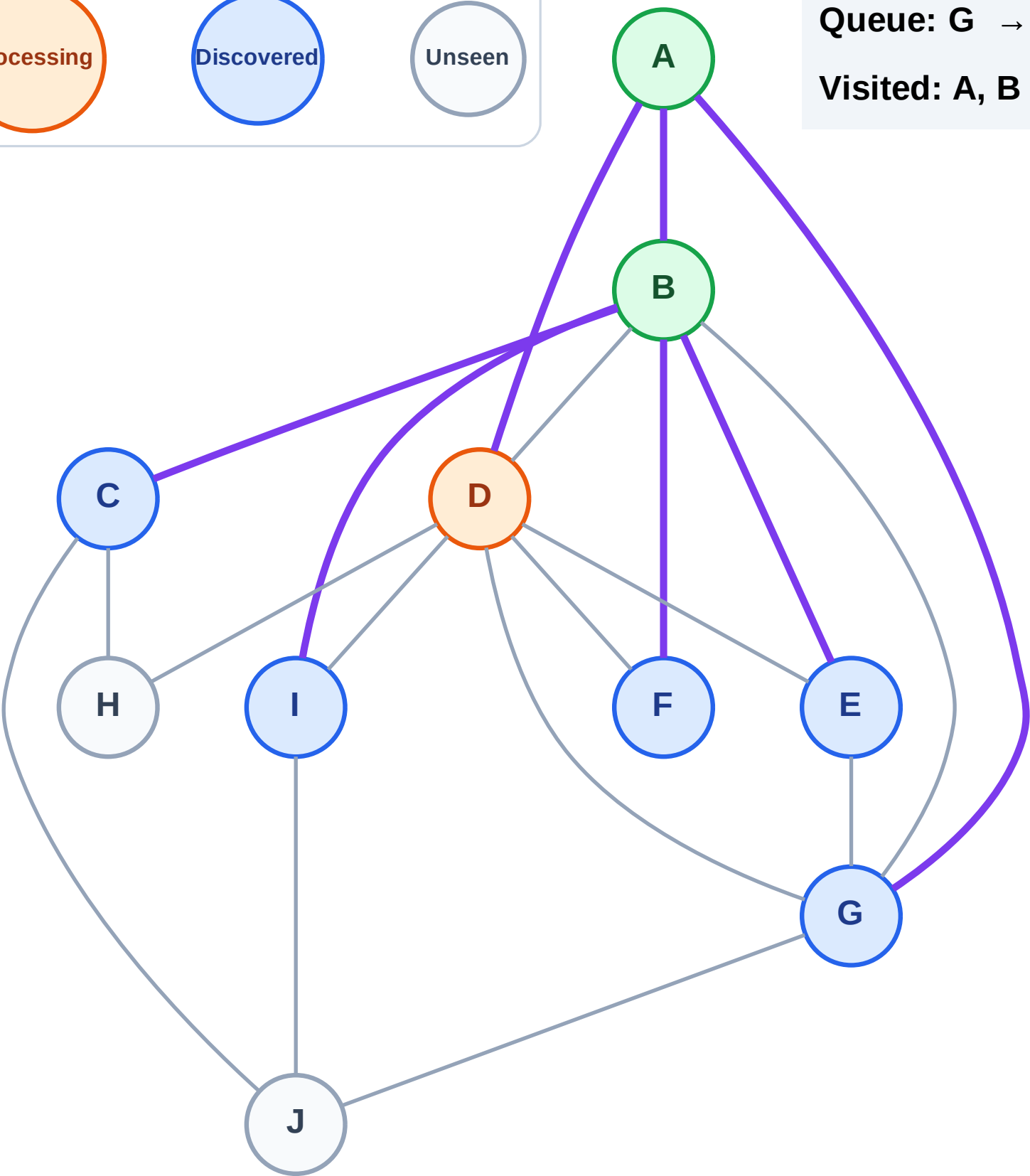
Complete

Processing

Discovered

Unseen

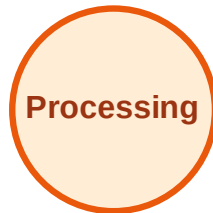
Queue: G → C → E → F → I
Visited: A, B



BFS Traversal

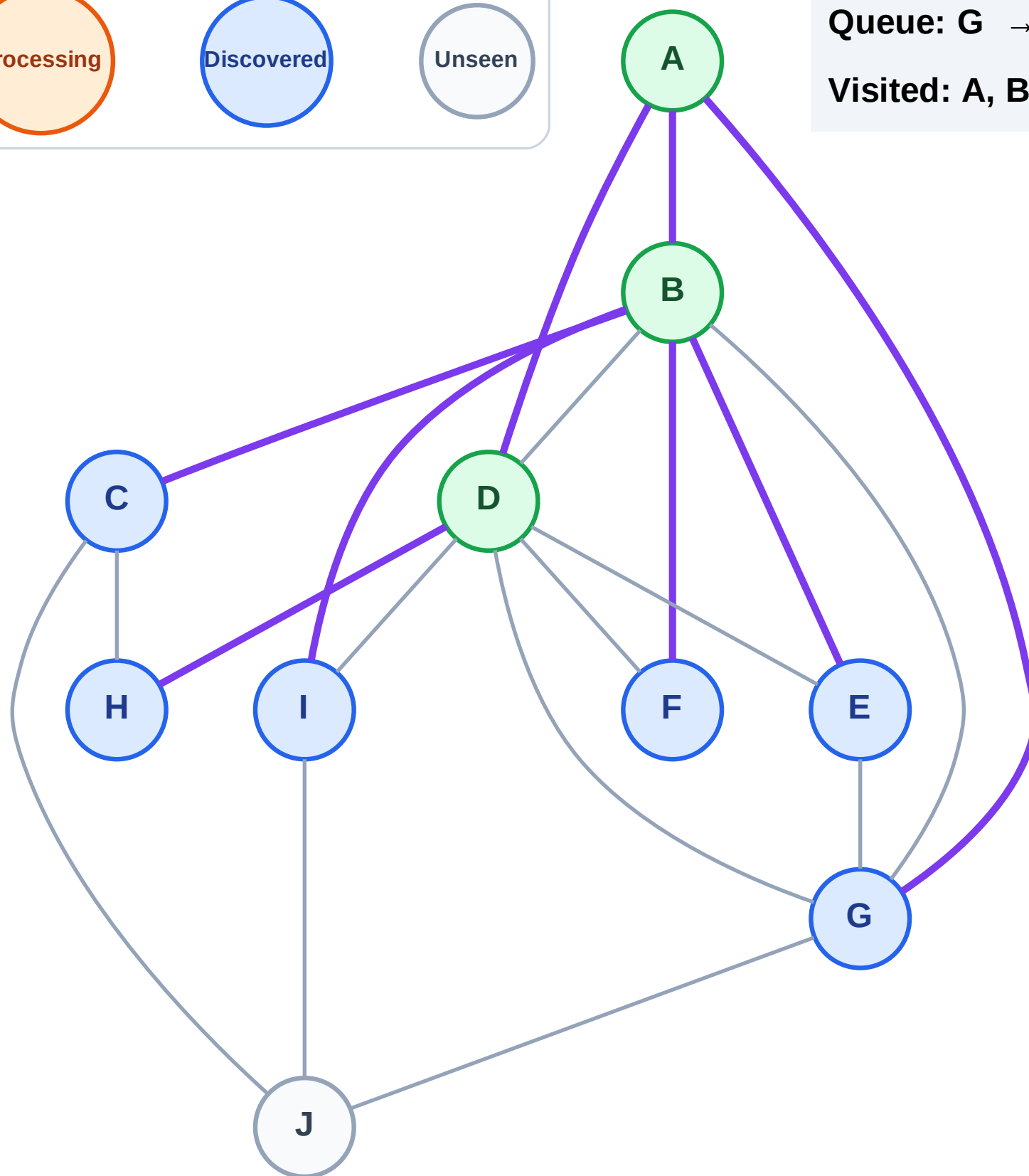
Step 7: Discover H from D

Legend



Queue: G → C → E → F → I → H

Visited: A, B, D



BFS Traversal

Step 8: Process node G

Legend

Complete

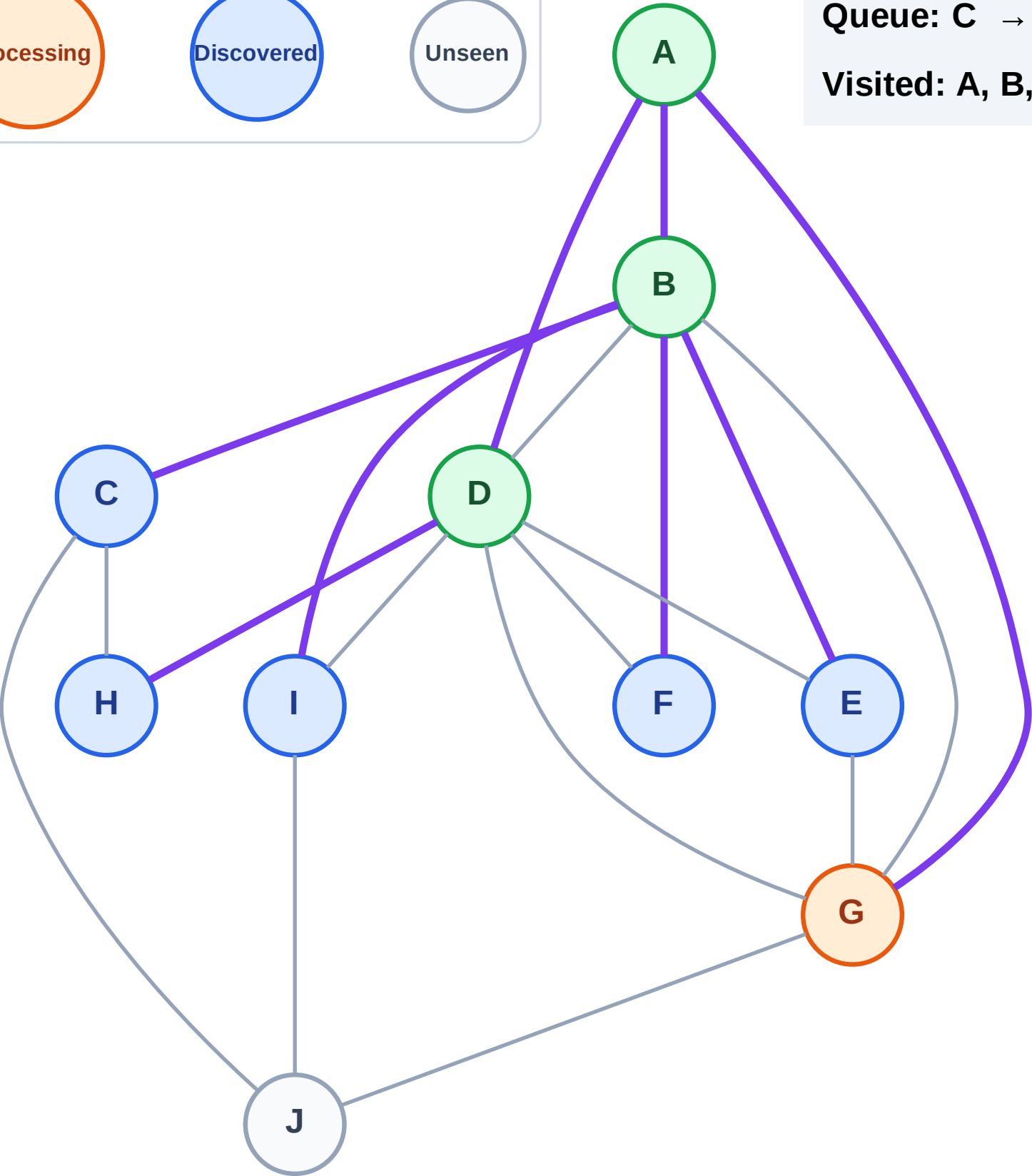
Processing

Discovered

Unseen

Queue: C → E → F → I → H

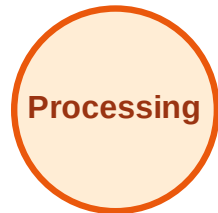
Visited: A, B, D



BFS Traversal

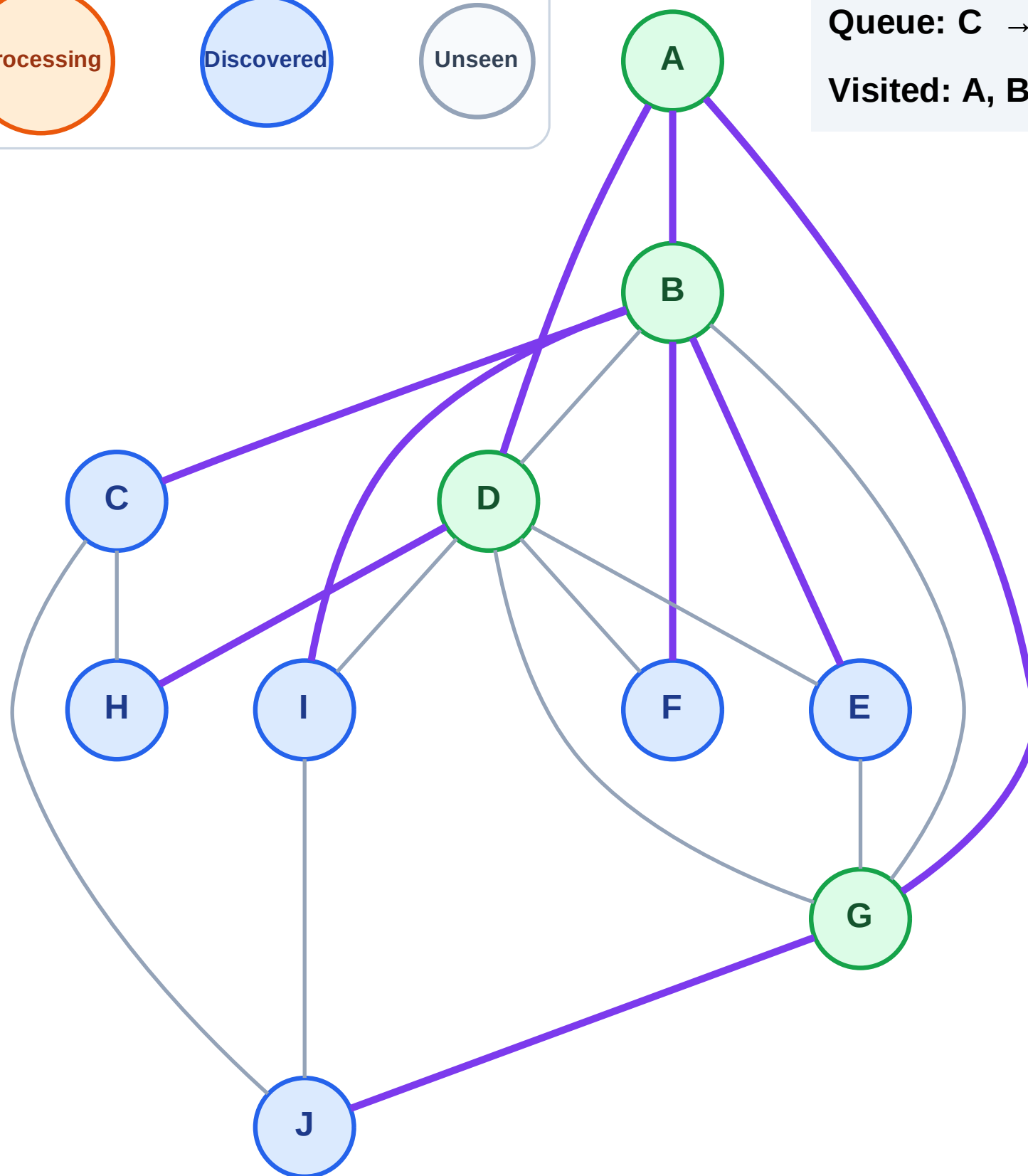
Step 9: Discover J from G

Legend



Queue: C → E → F → I → H → J

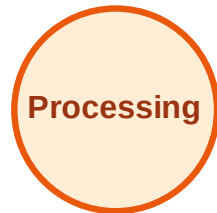
Visited: A, B, D, G



BFS Traversal

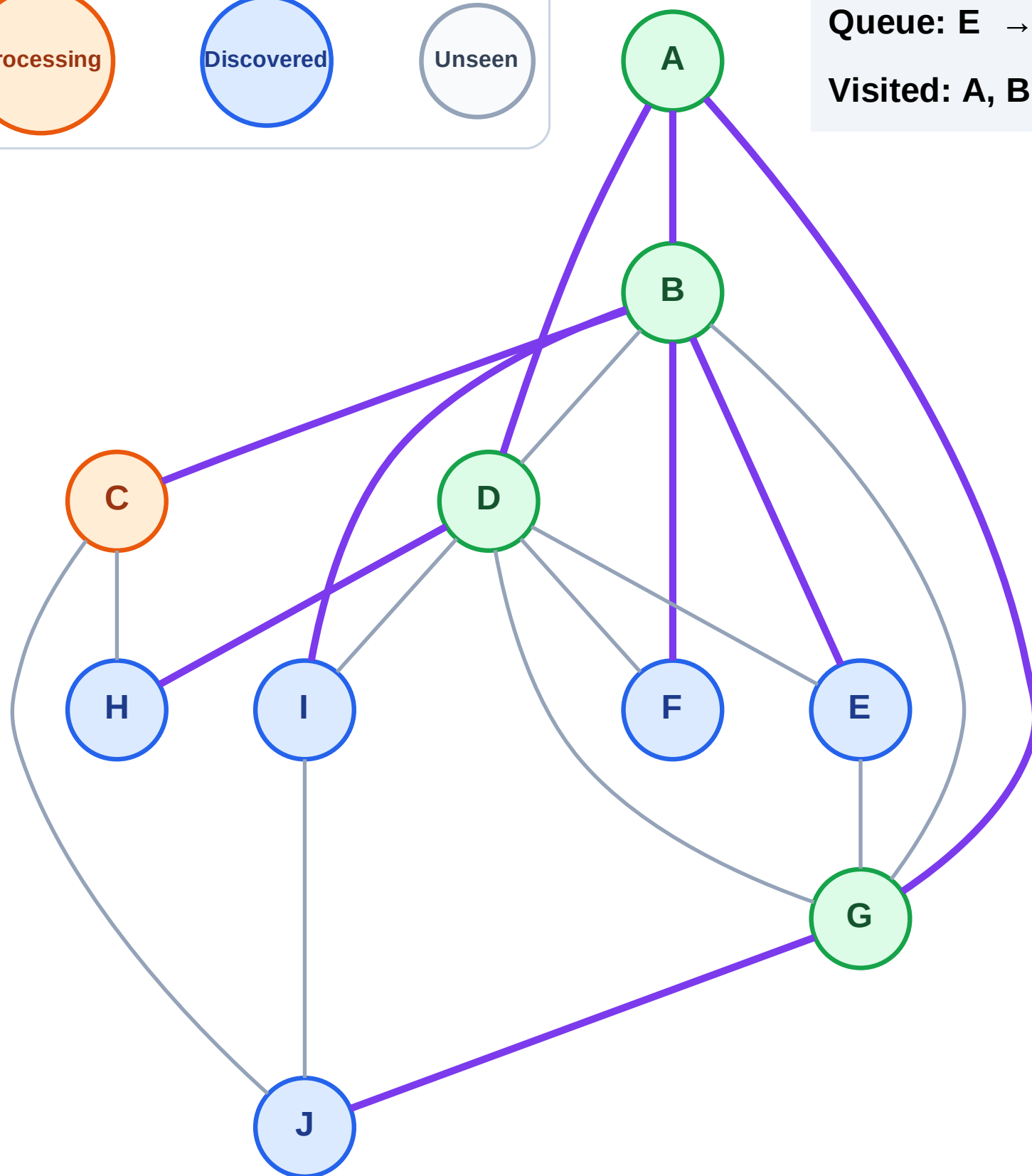
Step 10: Process node C

Legend



Queue: E → F → I → H → J

Visited: A, B, D, G



BFS Traversal

Step 11: No new neighbors from C

Legend

Complete

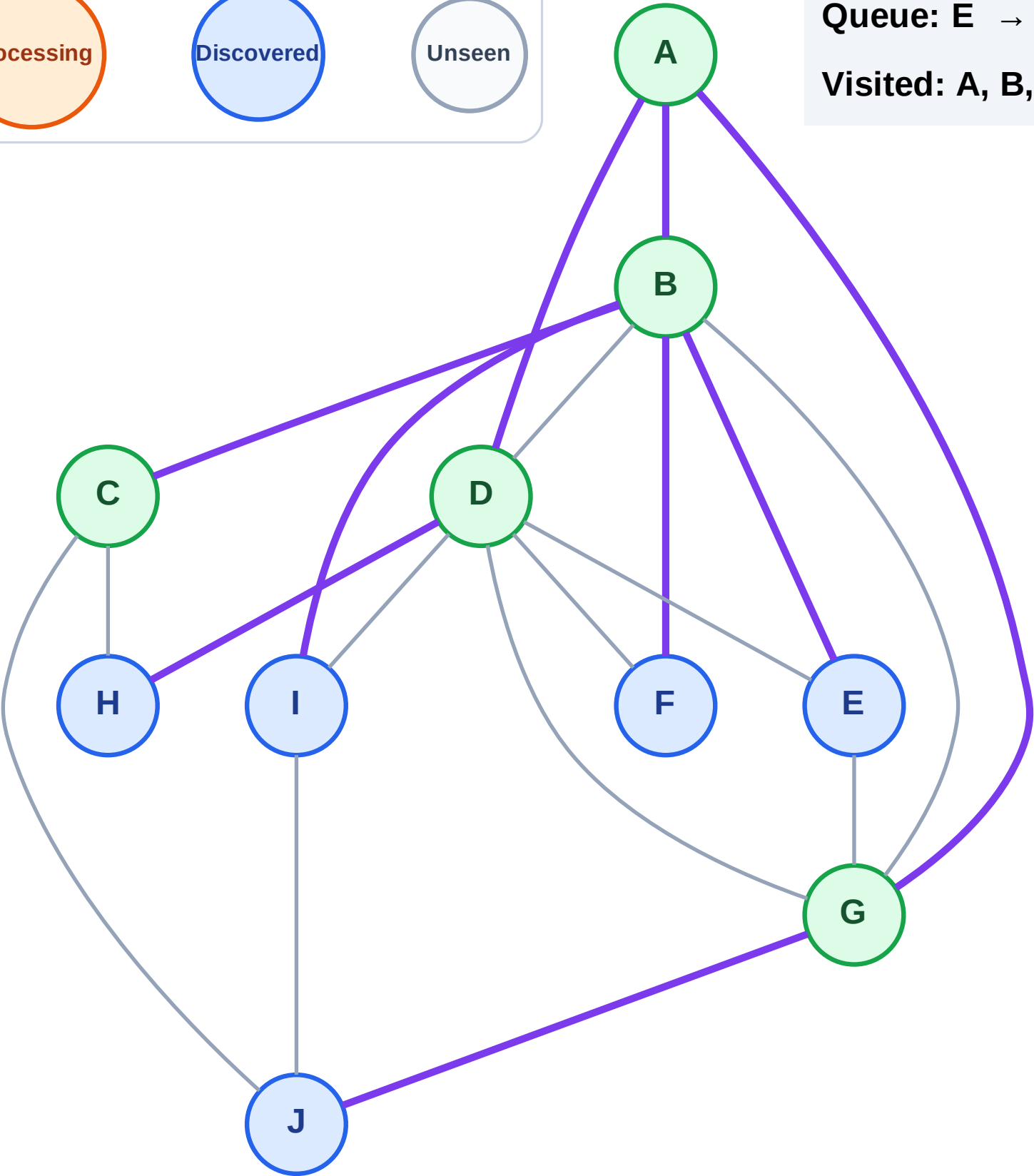
Processing

Discovered

Unseen

Queue: E → F → I → H → J

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node E

Legend

Complete

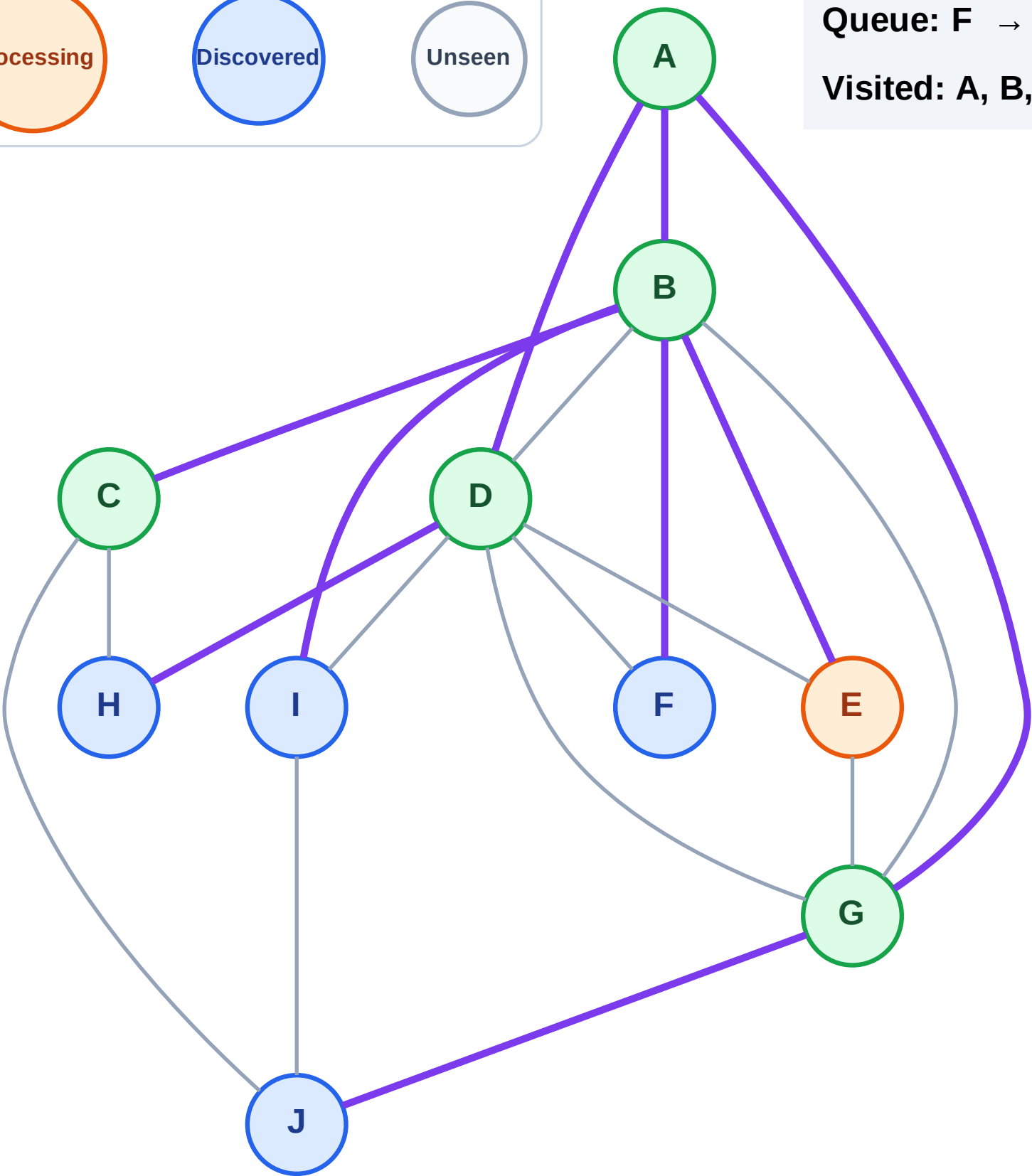
Processing

Discovered

Unseen

Queue: F → I → H → J

Visited: A, B, C, D, G



BFS Traversal

Step 13: No new neighbors from E

Legend

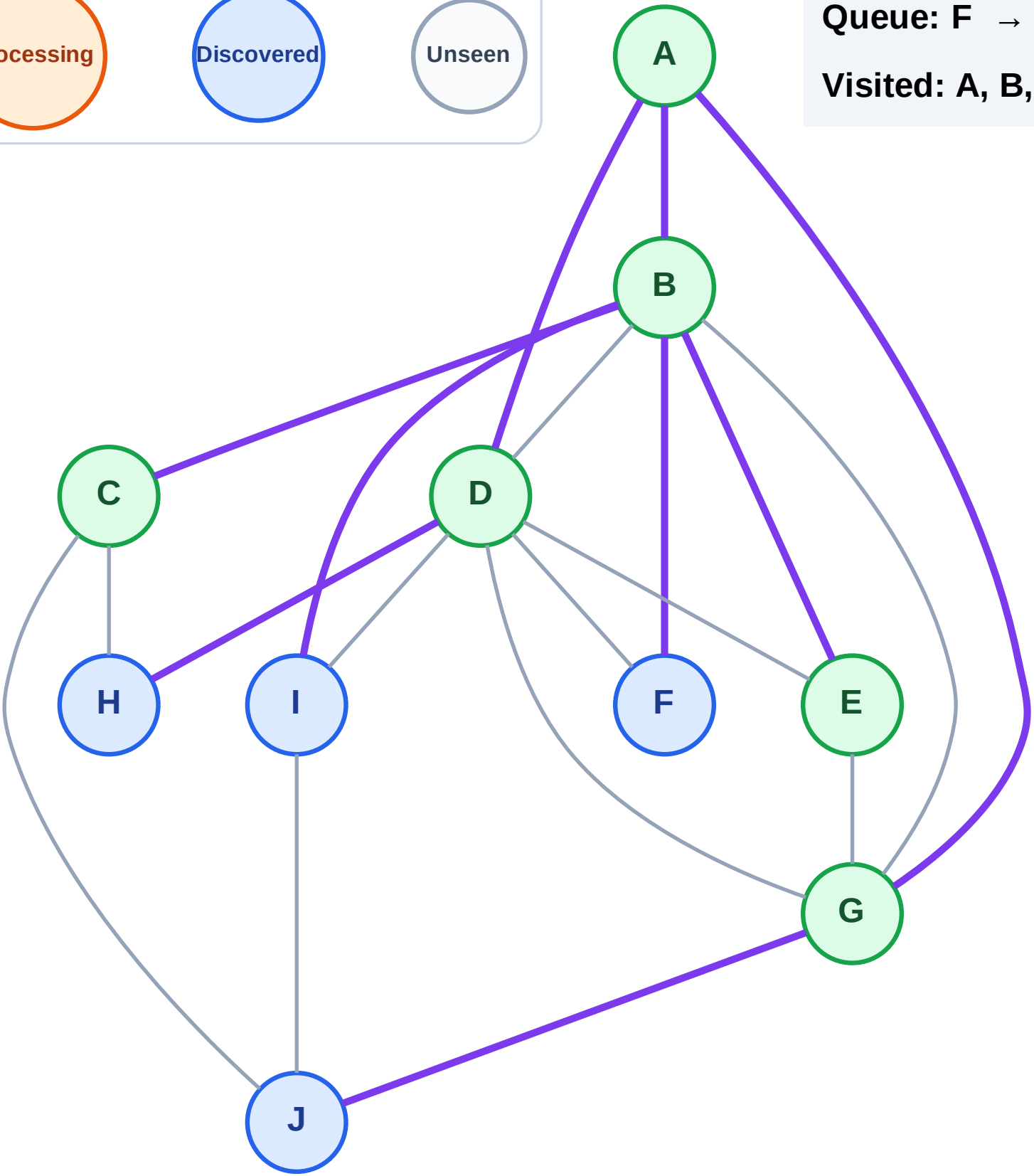
Complete

Processing

Discovered

Unseen

Queue: F → I → H → J
Visited: A, B, C, D, E, G



BFS Traversal

Step 14: Process node F

Legend

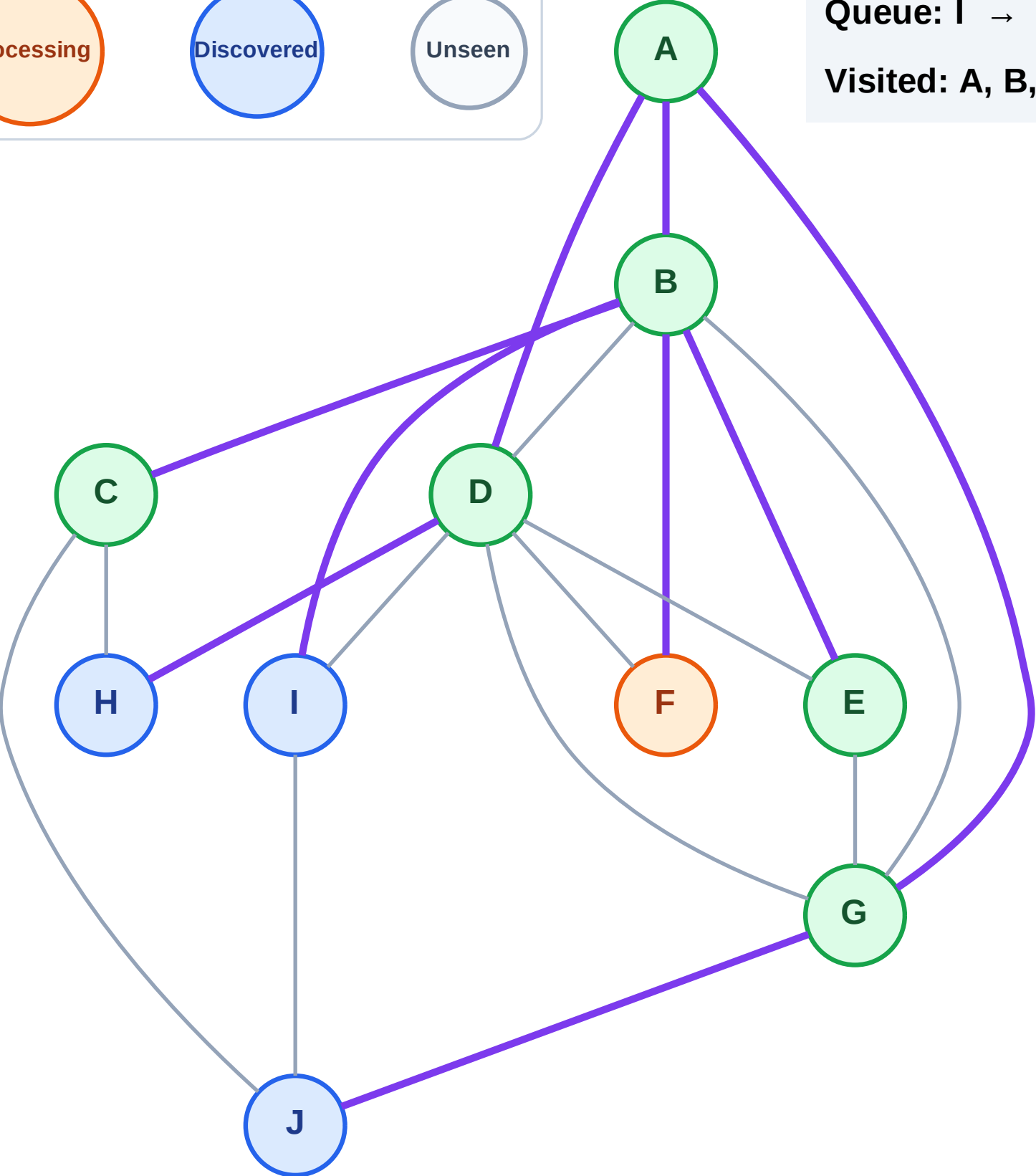
Complete

Processing

Discovered

Unseen

Queue: I → H → J
Visited: A, B, C, D, E, G



BFS Traversal

Step 15: No new neighbors from F

Legend

Complete

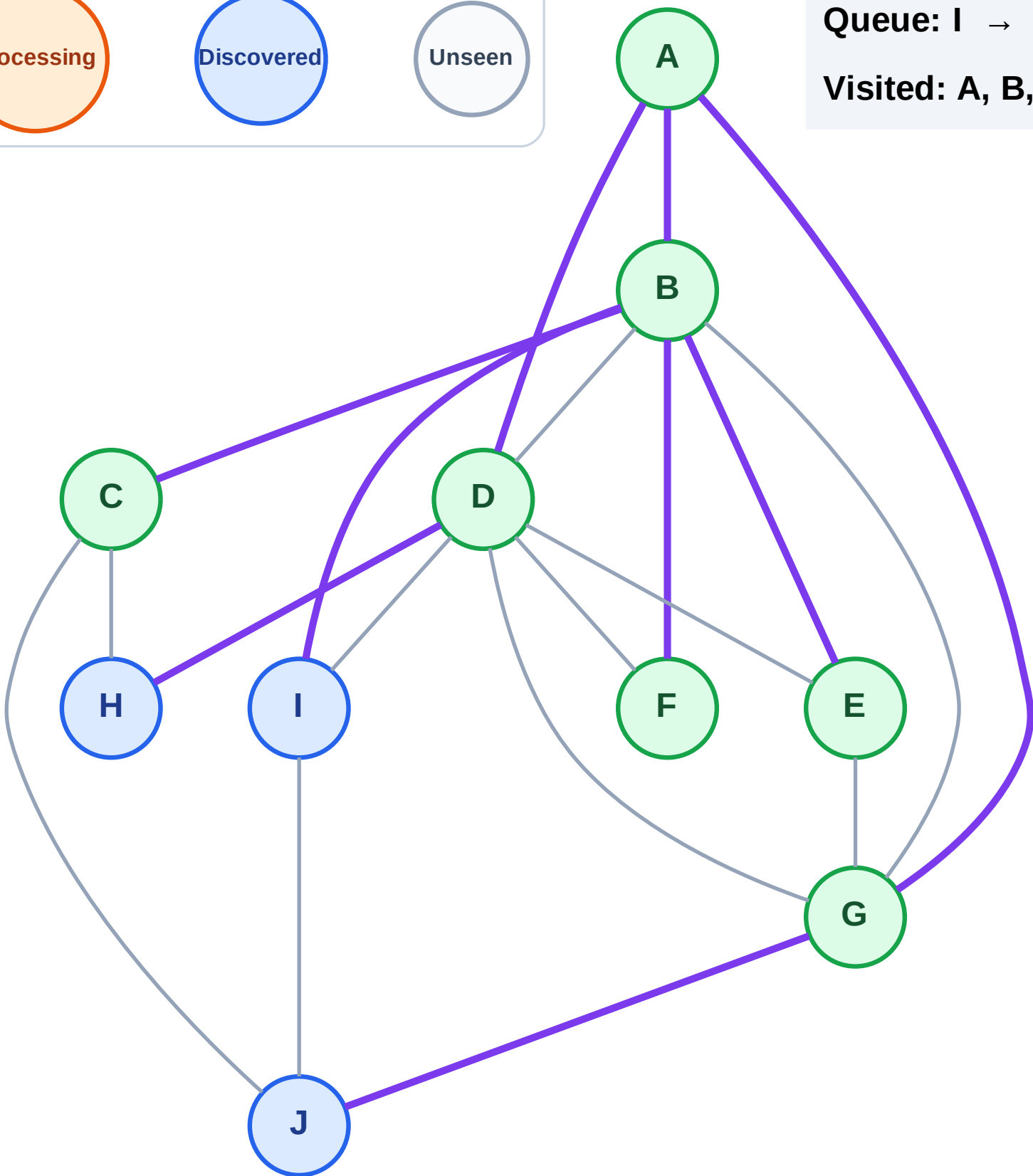
Processing

Discovered

Unseen

Queue: I → H → J

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 16: Process node I

Legend

Complete

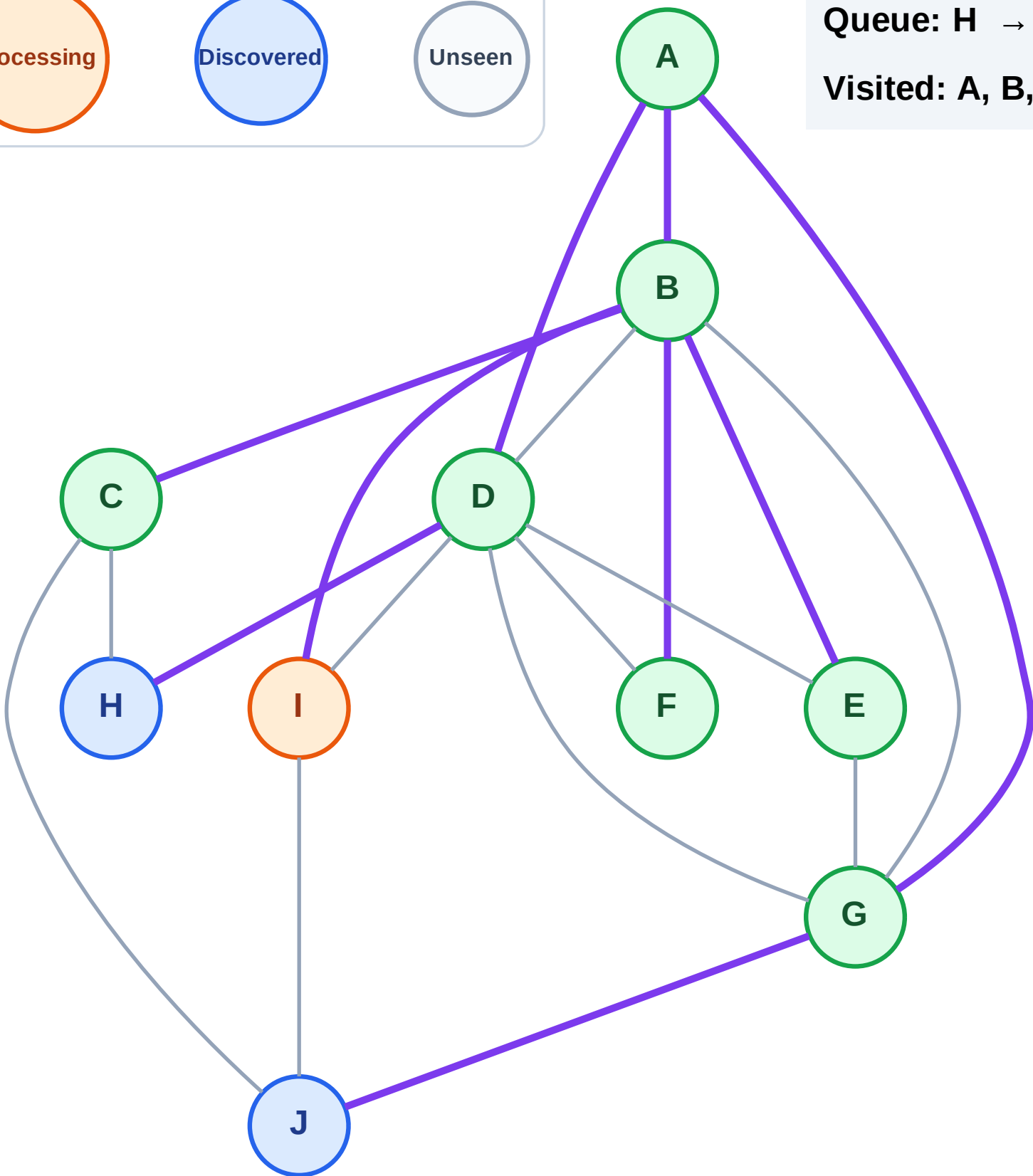
Processing

Discovered

Unseen

Queue: H → J

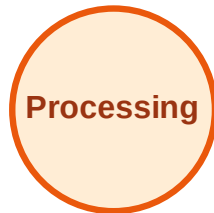
Visited: A, B, C, D, E, F, G



BFS Traversal

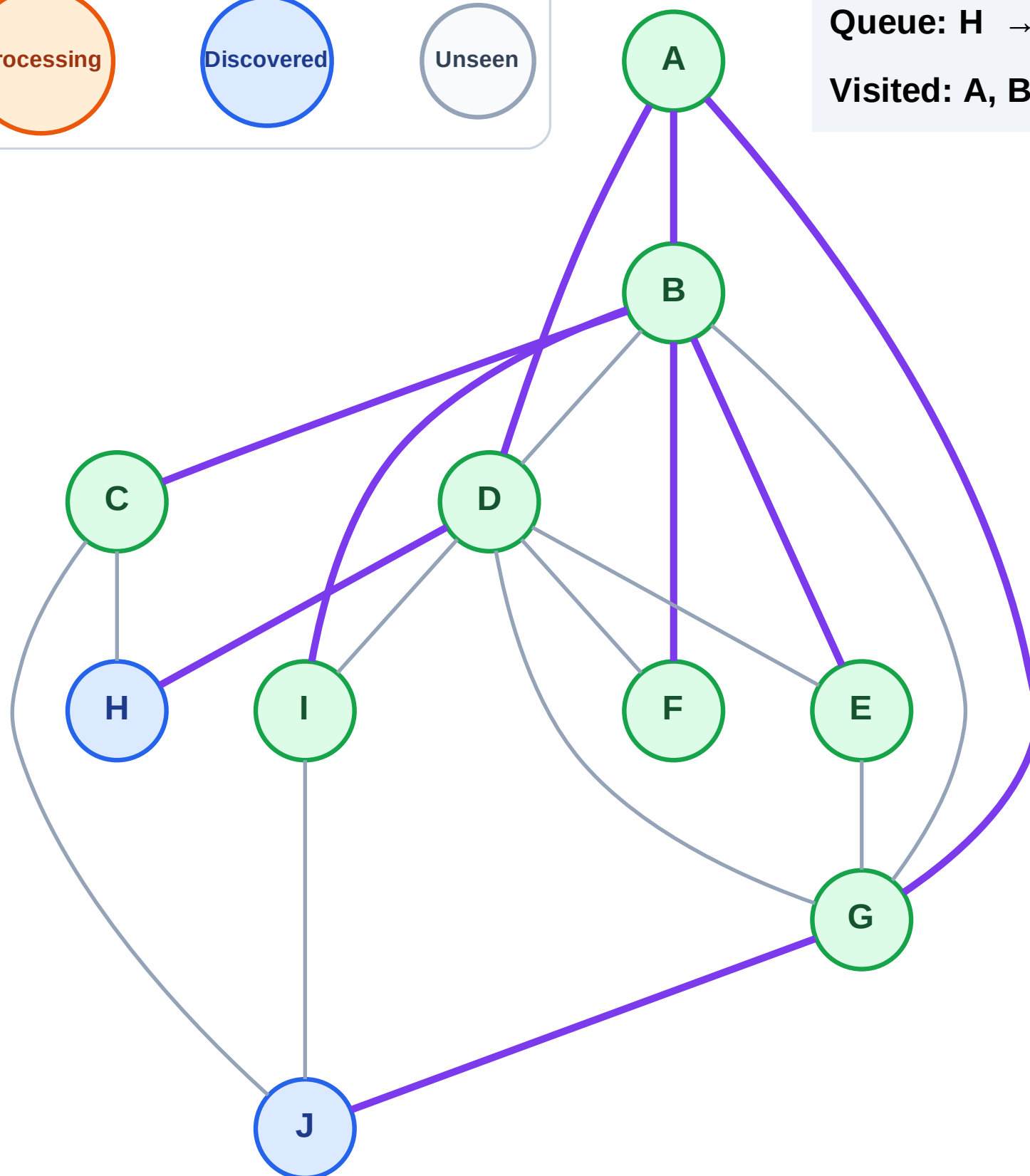
Step 17: No new neighbors from I

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

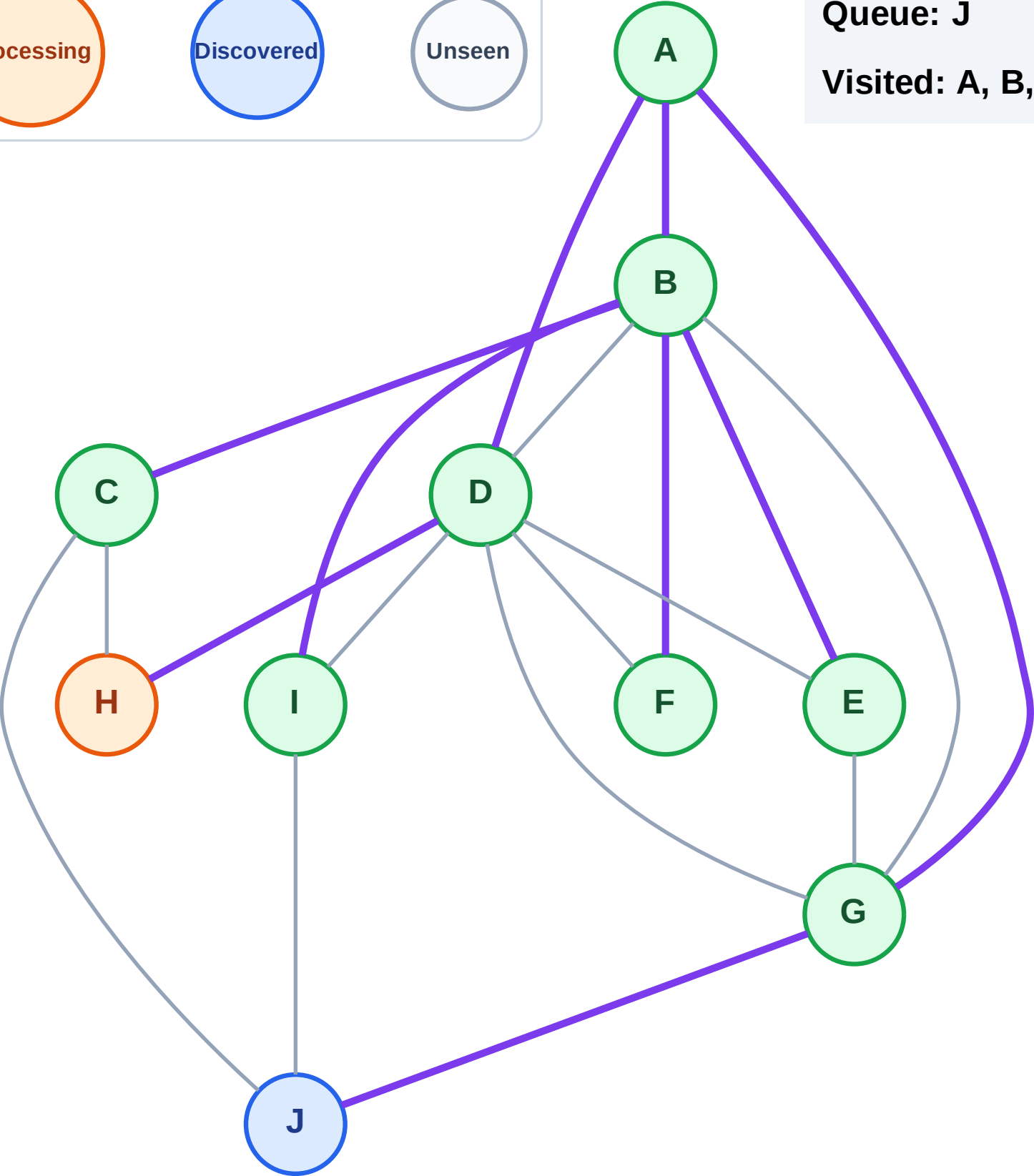
Processing

Discovered

Unseen

Queue: J

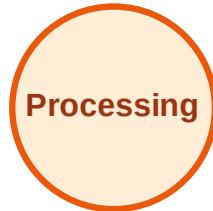
Visited: A, B, C, D, E, F, G, I



BFS Traversal

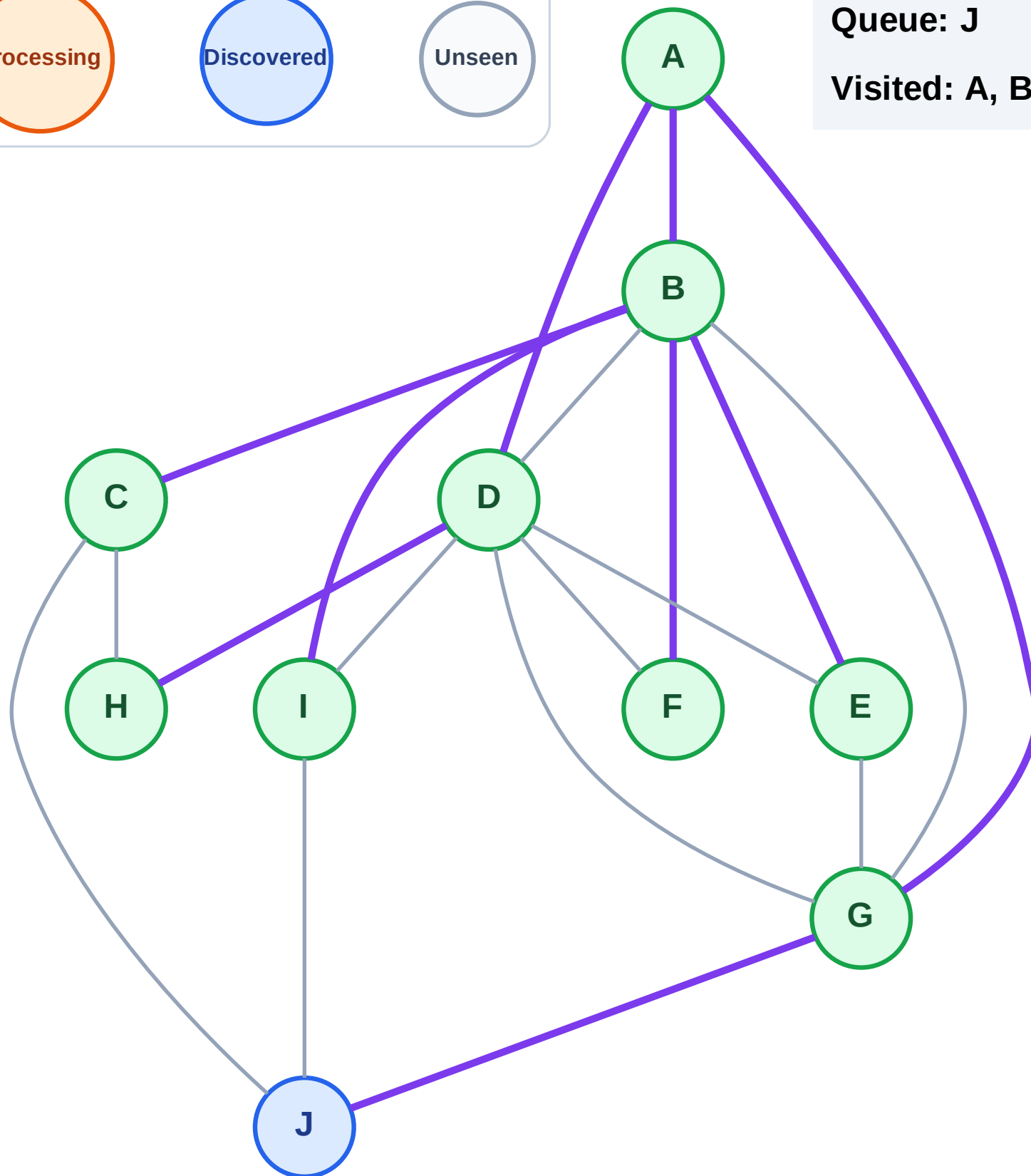
Step 19: No new neighbors from H

Legend



Queue: J

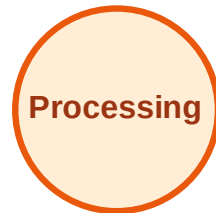
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

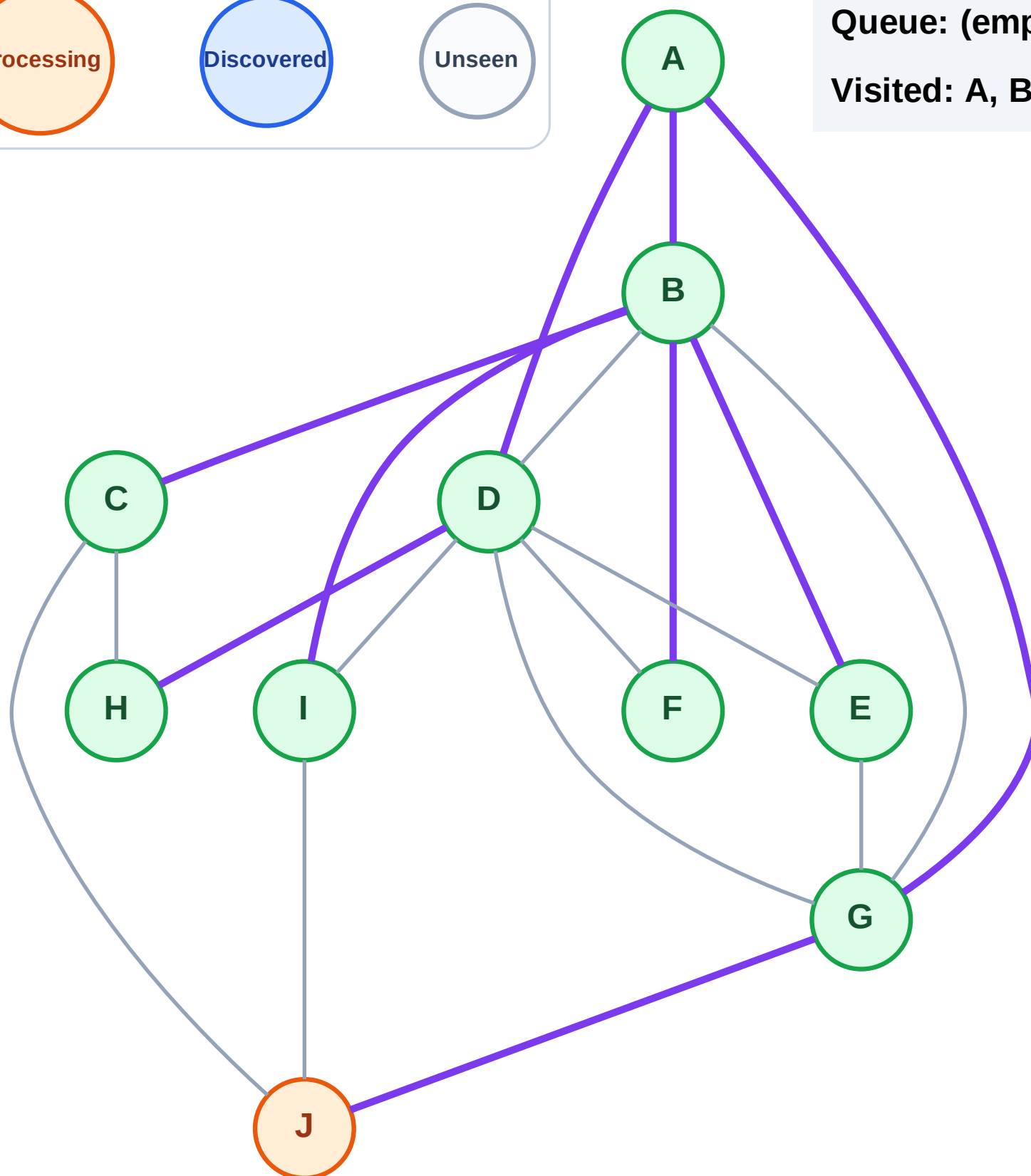
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



Step 21: No new neighbors from J

Legend



The diagram shows a graph with 10 nodes (A-J) and their connections. The nodes are colored based on their state: A, B, C, D, E, F, G, H, and I are light green with green borders, indicating they are 'Discovered'. Node J is light blue with a blue border, indicating it is 'Unseen'. The edges are colored based on their state: purple edges represent 'Processing' (A-B, A-G, B-D, B-E, C-D, D-H, D-I, D-F, E-G, F-G, G-I, G-J, H-I, I-J), and gray edges represent 'Unseen' (A-C, B-F, C-H, D-E, E-F, F-I, G-H, H-J, I-F, J-F).

Legend:

- Processing (orange circle)
- Discovered (blue circle)
- Unseen (gray circle)

Queue: (empty)
Visited: A, B

DFS Traversal

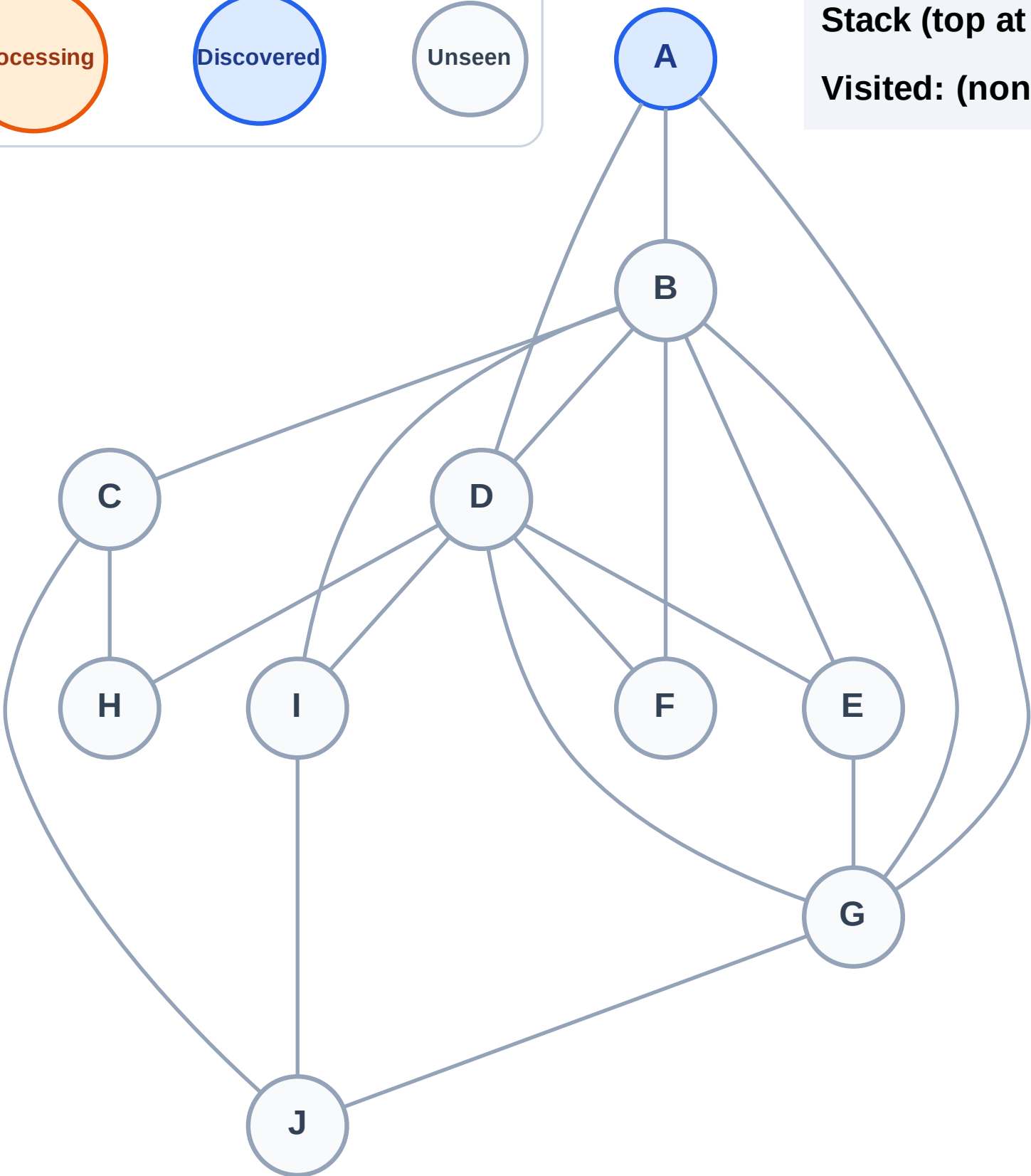
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

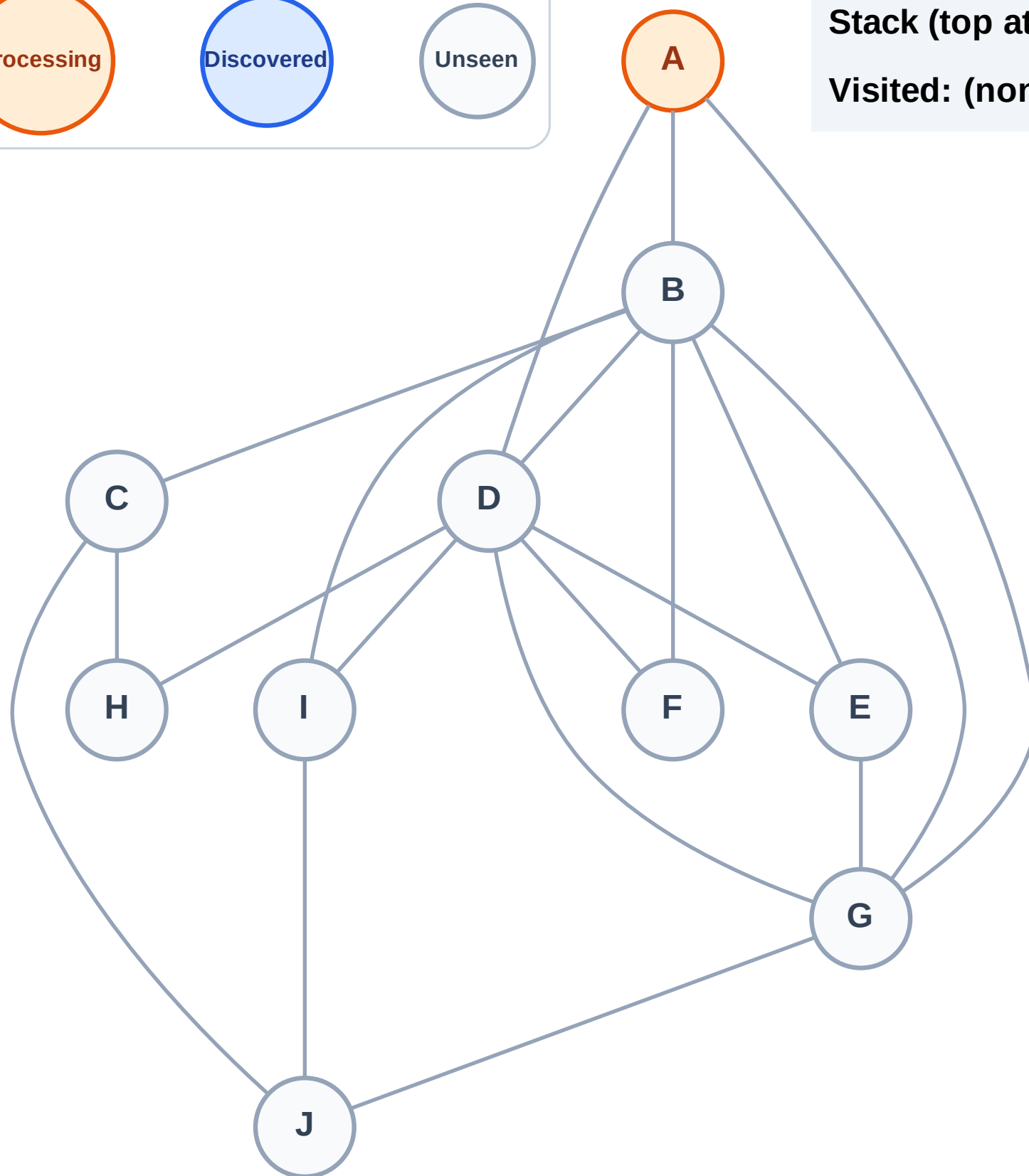
Processing

Discovered

Unseen

Stack (top at right): (empty)

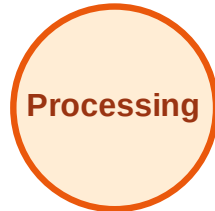
Visited: (none)



DFS Traversal

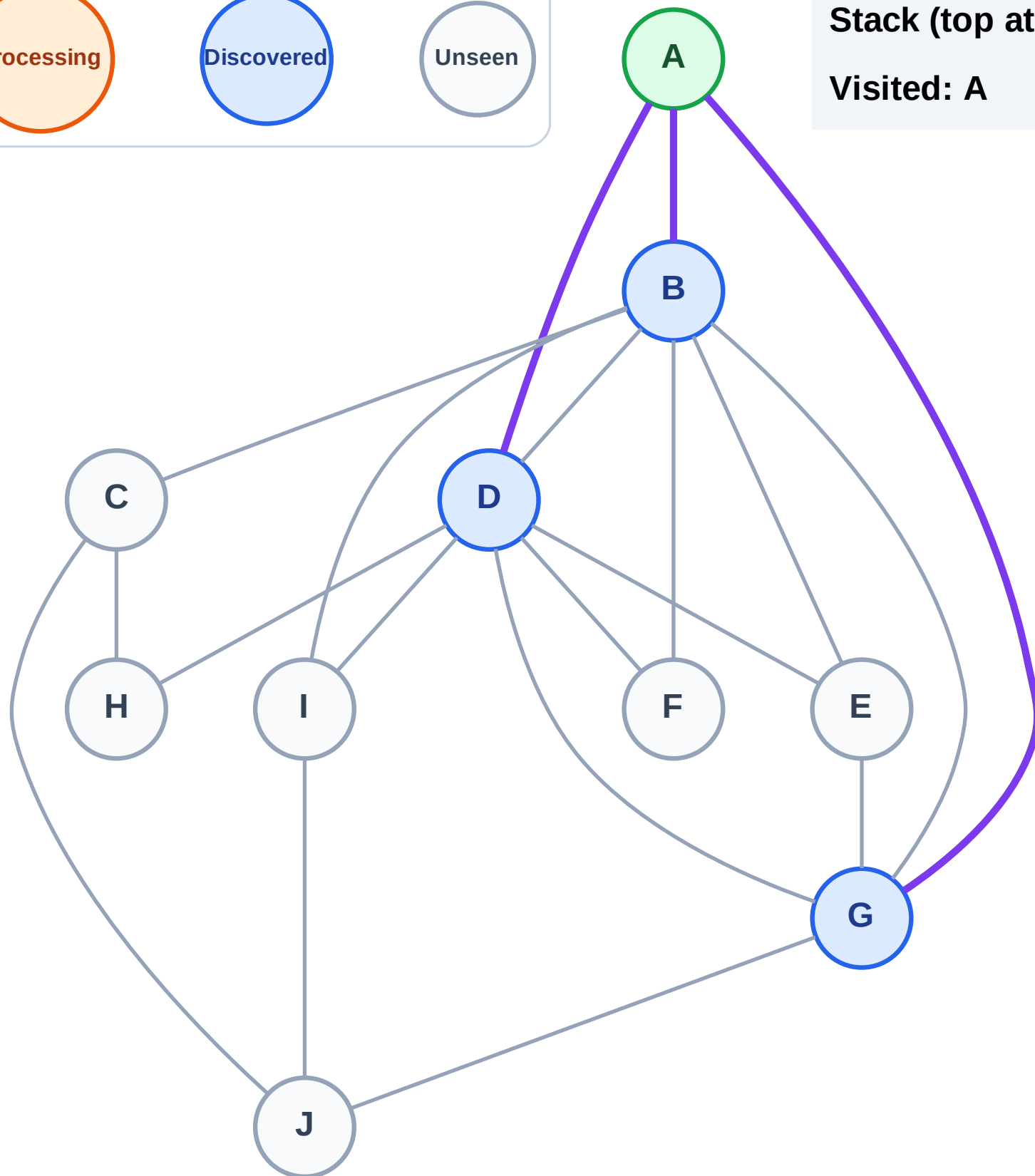
Step 3: Discover G, D, B from A

Legend



Stack (top at right): G | D | B

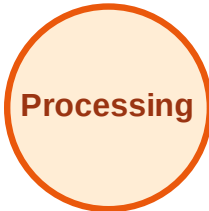
Visited: A



DFS Traversal

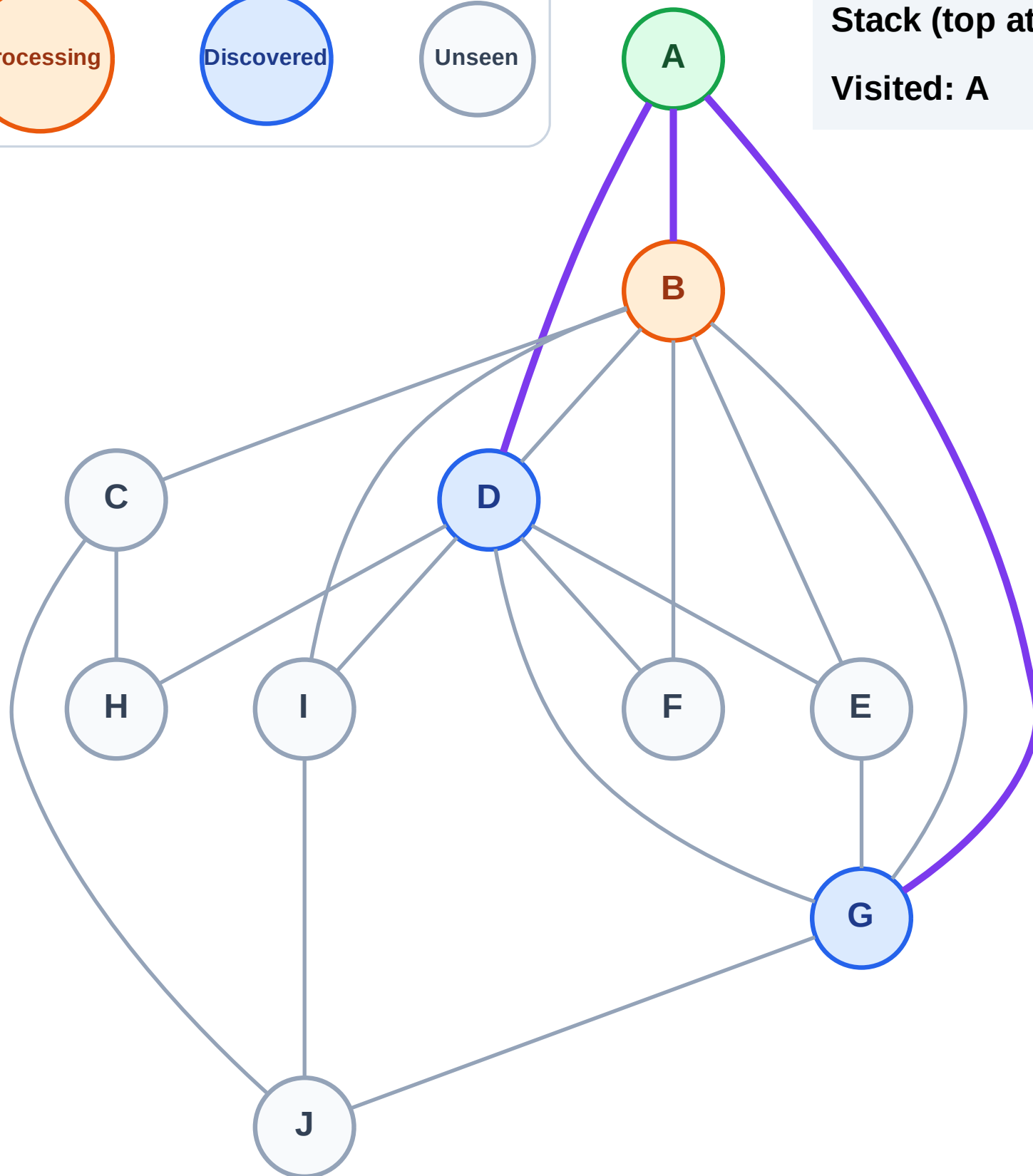
Step 4: Process node B

Legend



Stack (top at right): G | D

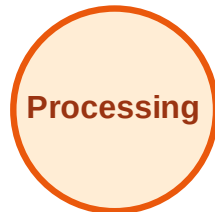
Visited: A



DFS Traversal

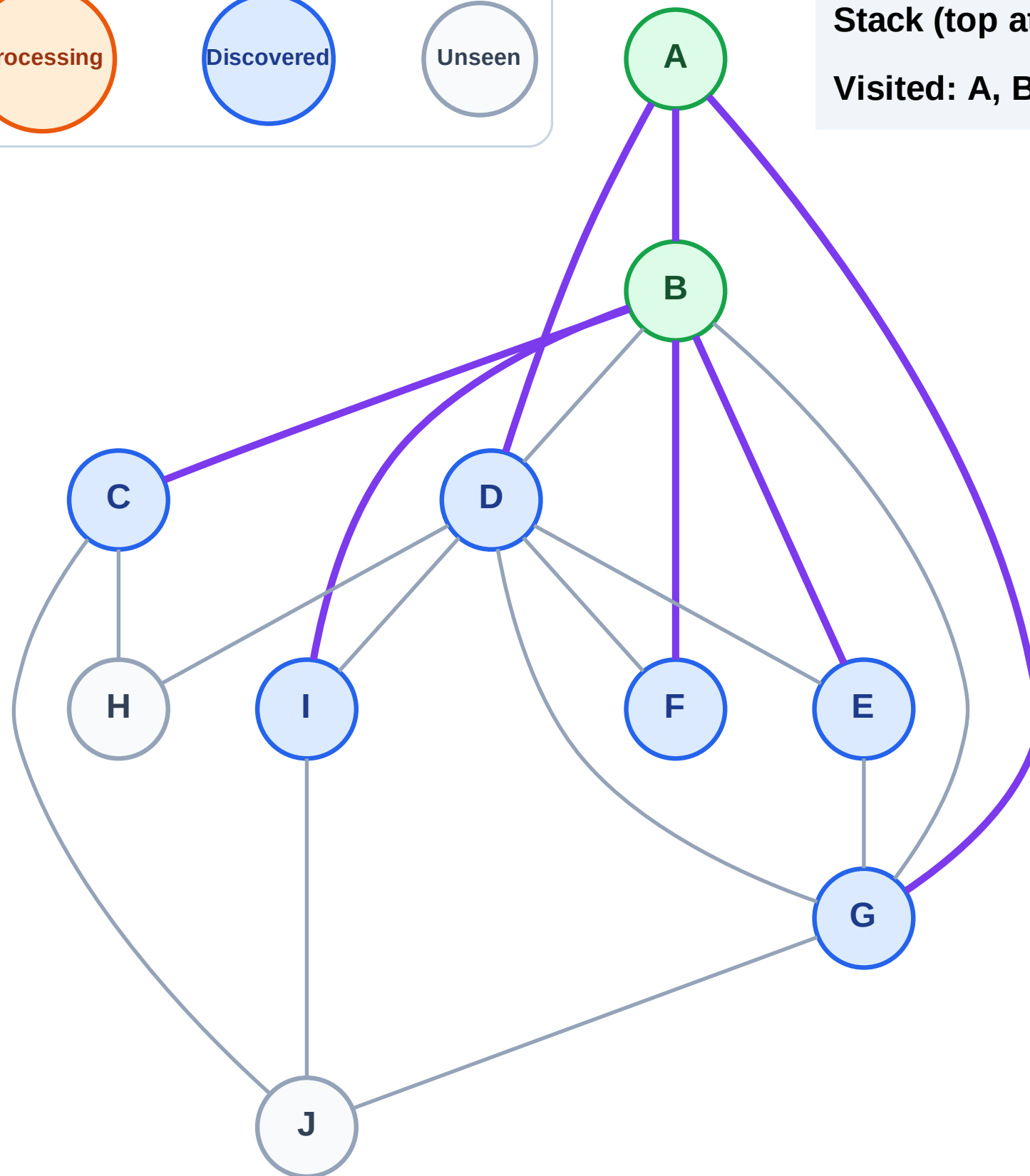
Step 5: Discover I, F, E, C from B

Legend



Stack (top at right): G | D | I | F | E | C

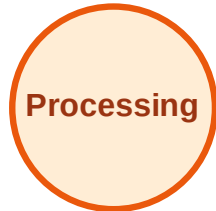
Visited: A, B



DFS Traversal

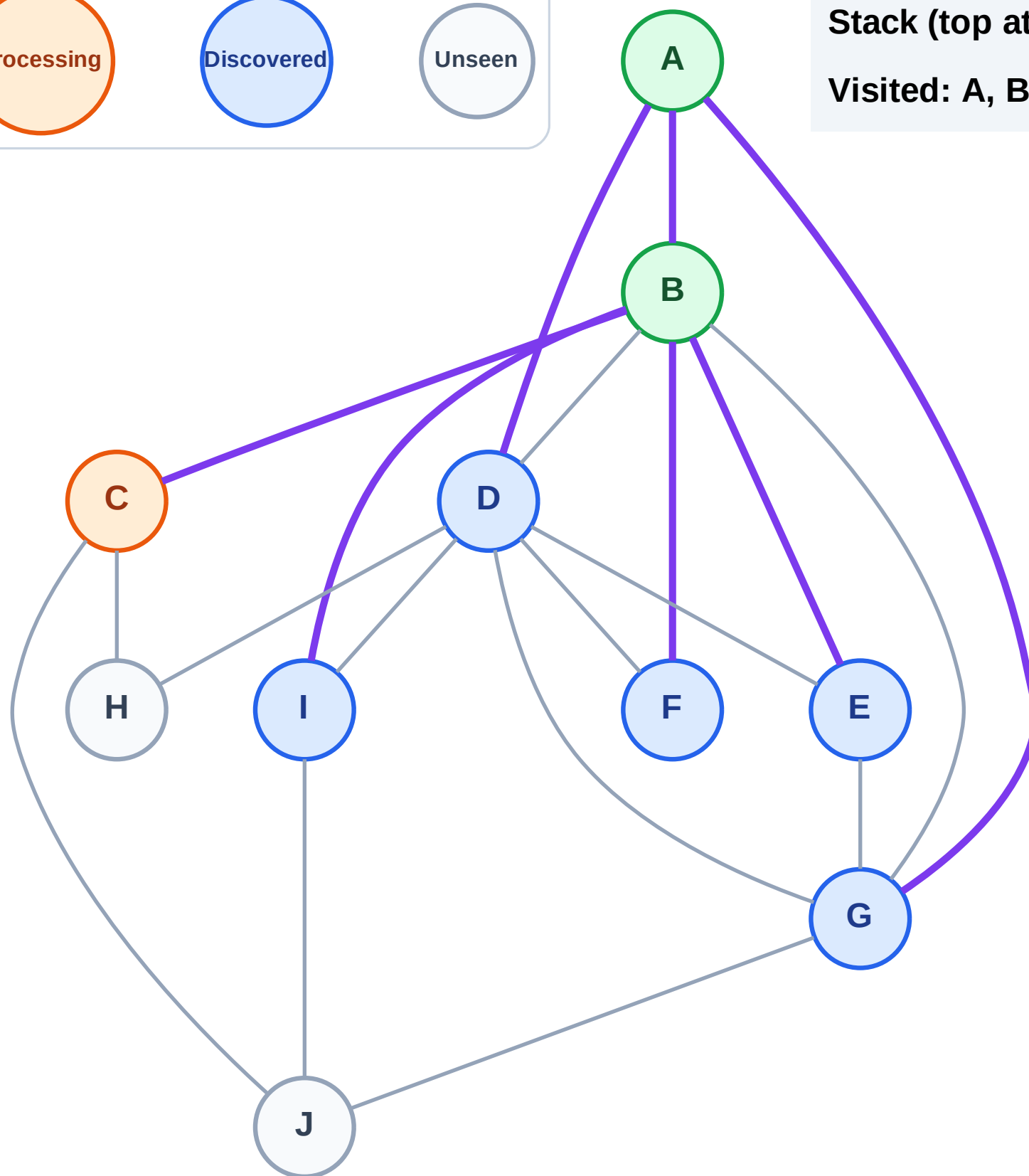
Step 6: Process node C

Legend



Stack (top at right): G | D | I | F | E

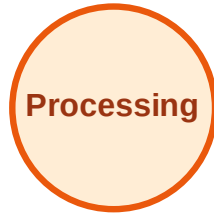
Visited: A, B



DFS Traversal

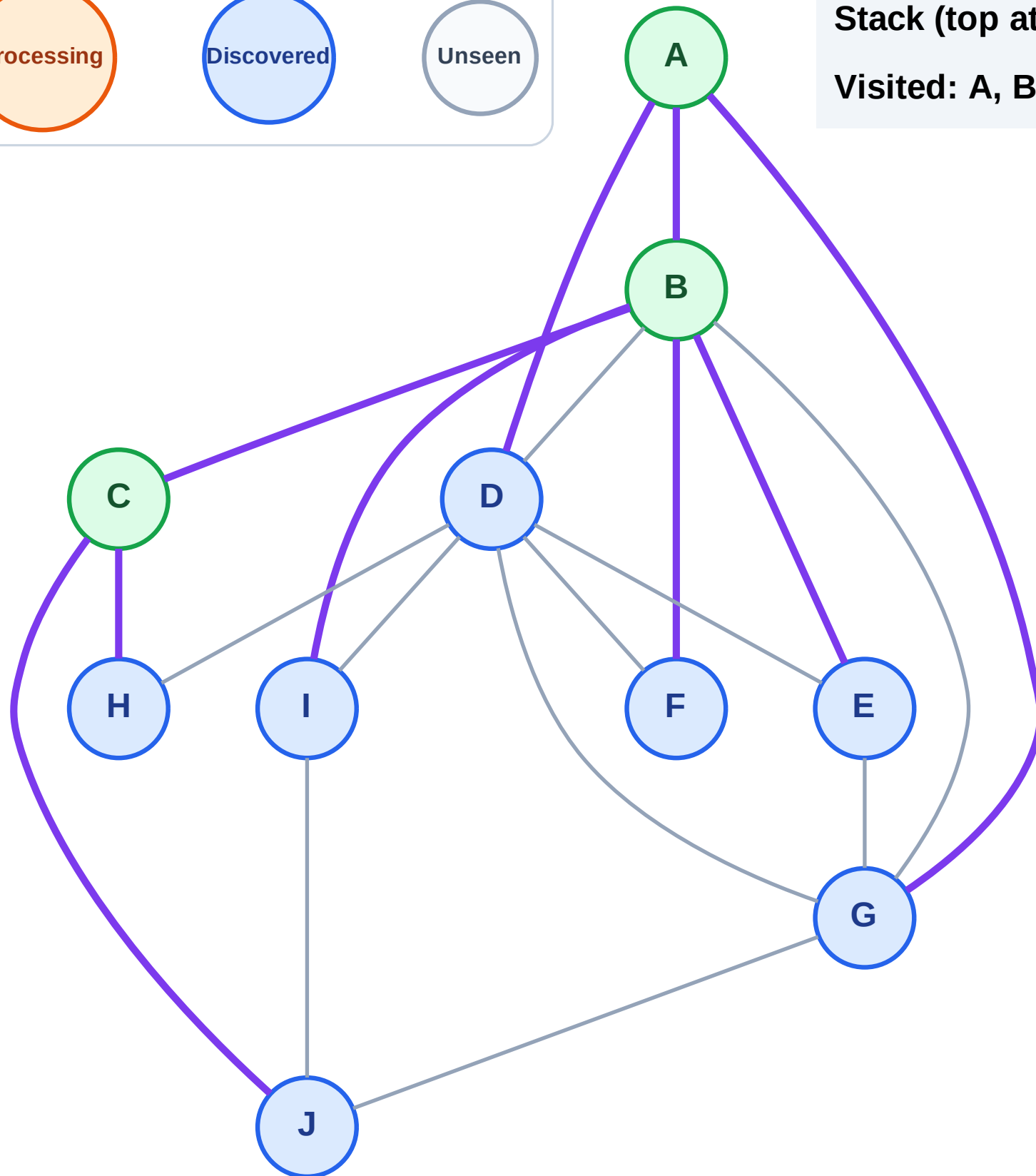
Step 7: Discover J, H from C

Legend



Stack (top at right): G | D | I | F | E | J | H

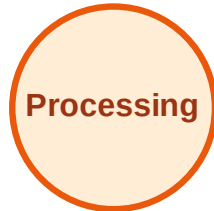
Visited: A, B, C



DFS Traversal

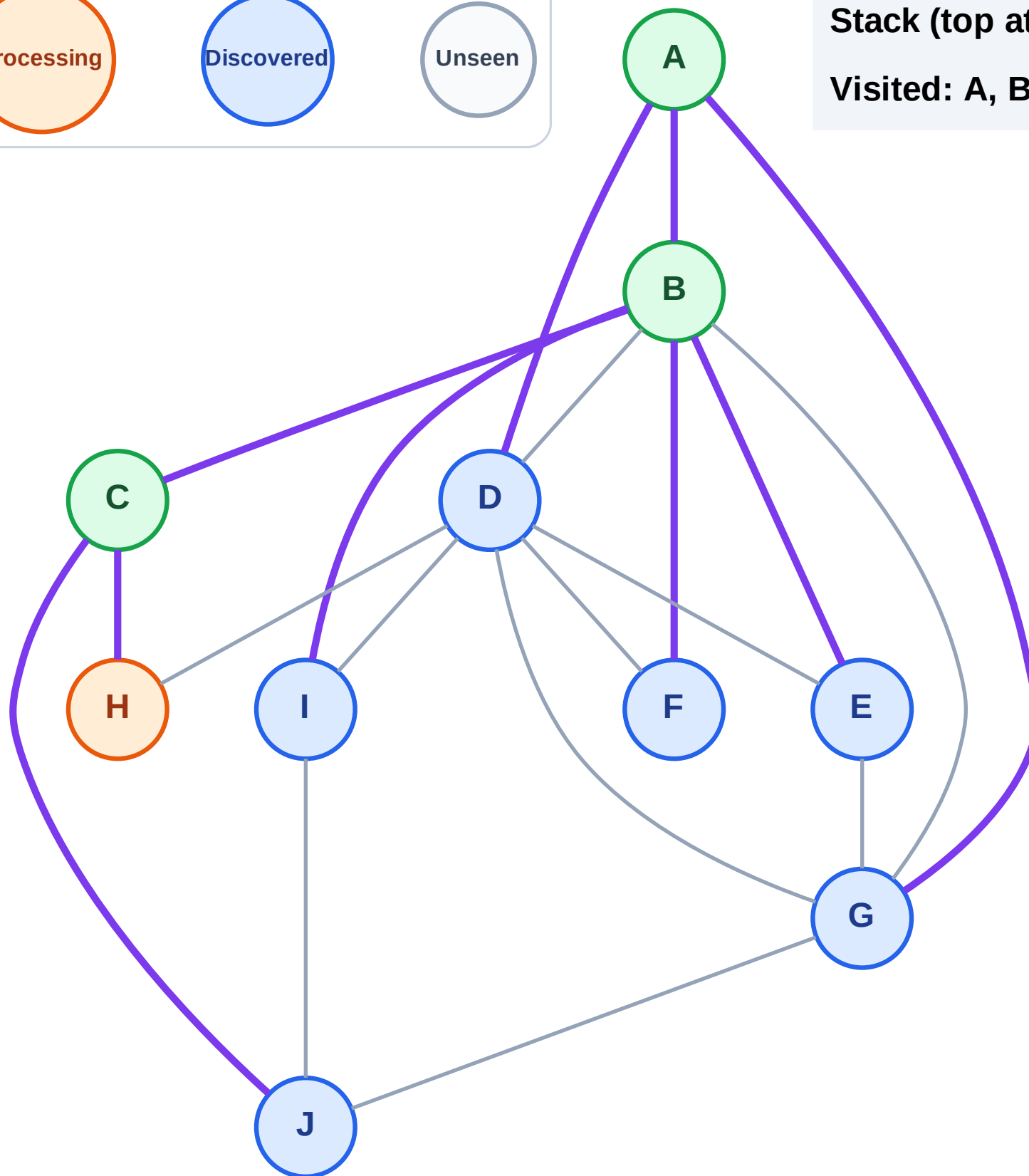
Step 8: Process node H

Legend



Stack (top at right): G | D | I | F | E | J

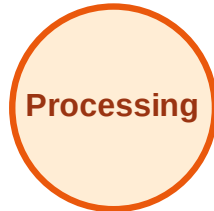
Visited: A, B, C



DFS Traversal

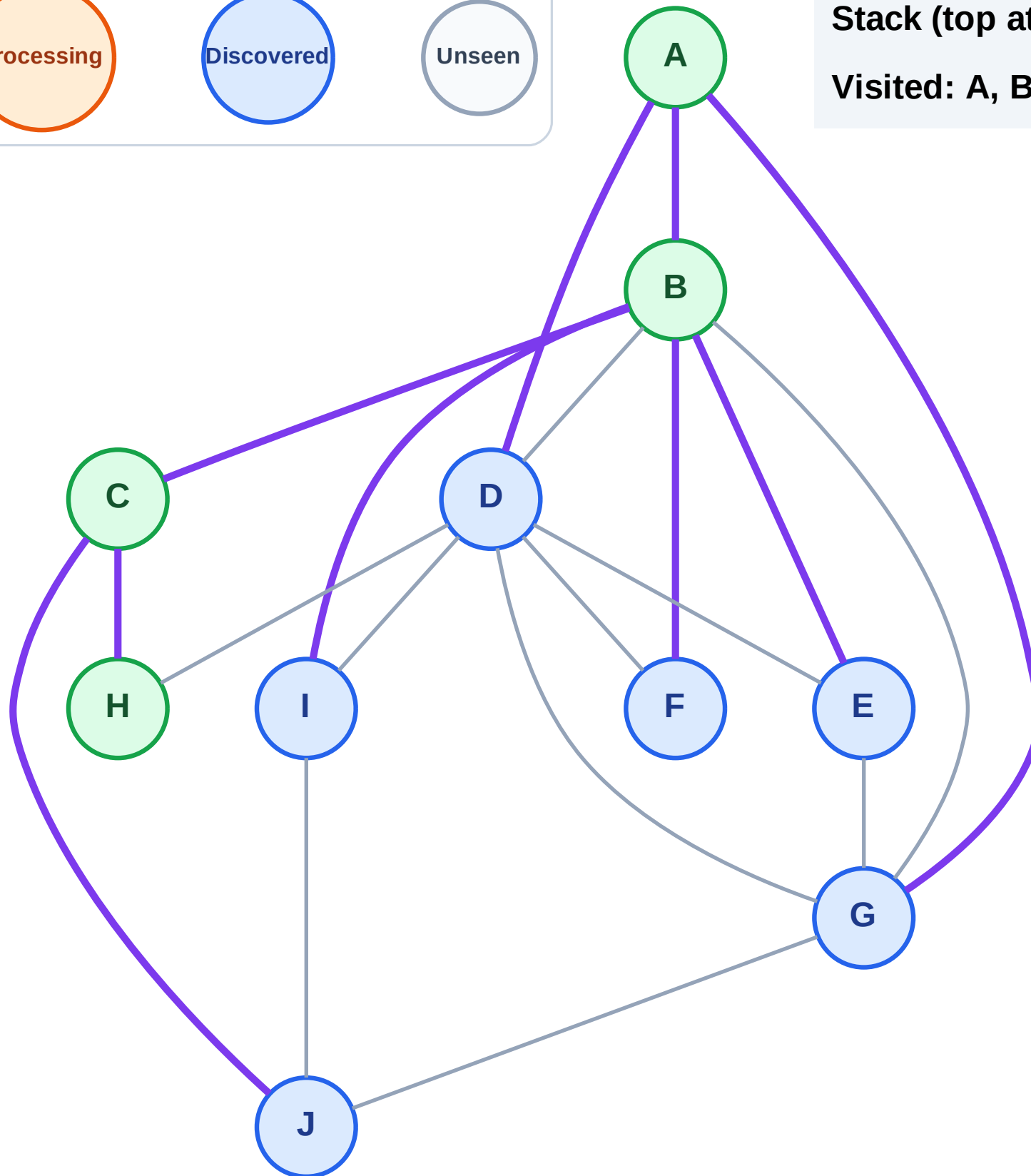
Step 9: No new neighbors from H

Legend



Stack (top at right): G | D | I | F | E | J

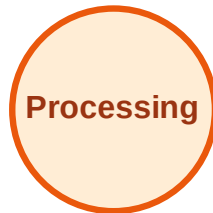
Visited: A, B, C, H



DFS Traversal

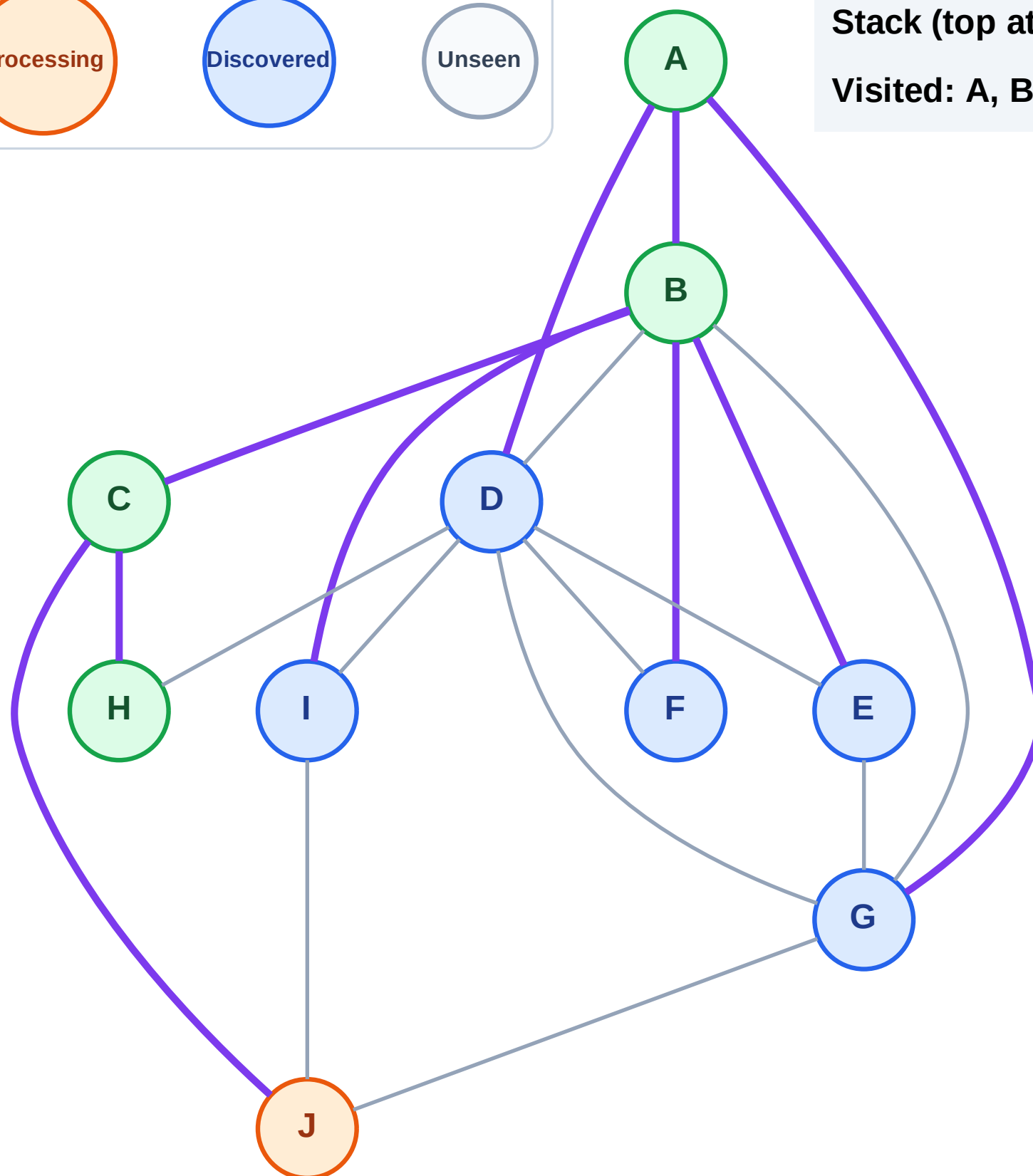
Step 10: Process node J

Legend



Stack (top at right): G | D | I | F | E

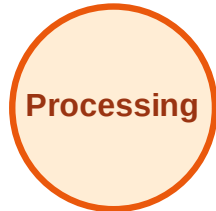
Visited: A, B, C, H



DFS Traversal

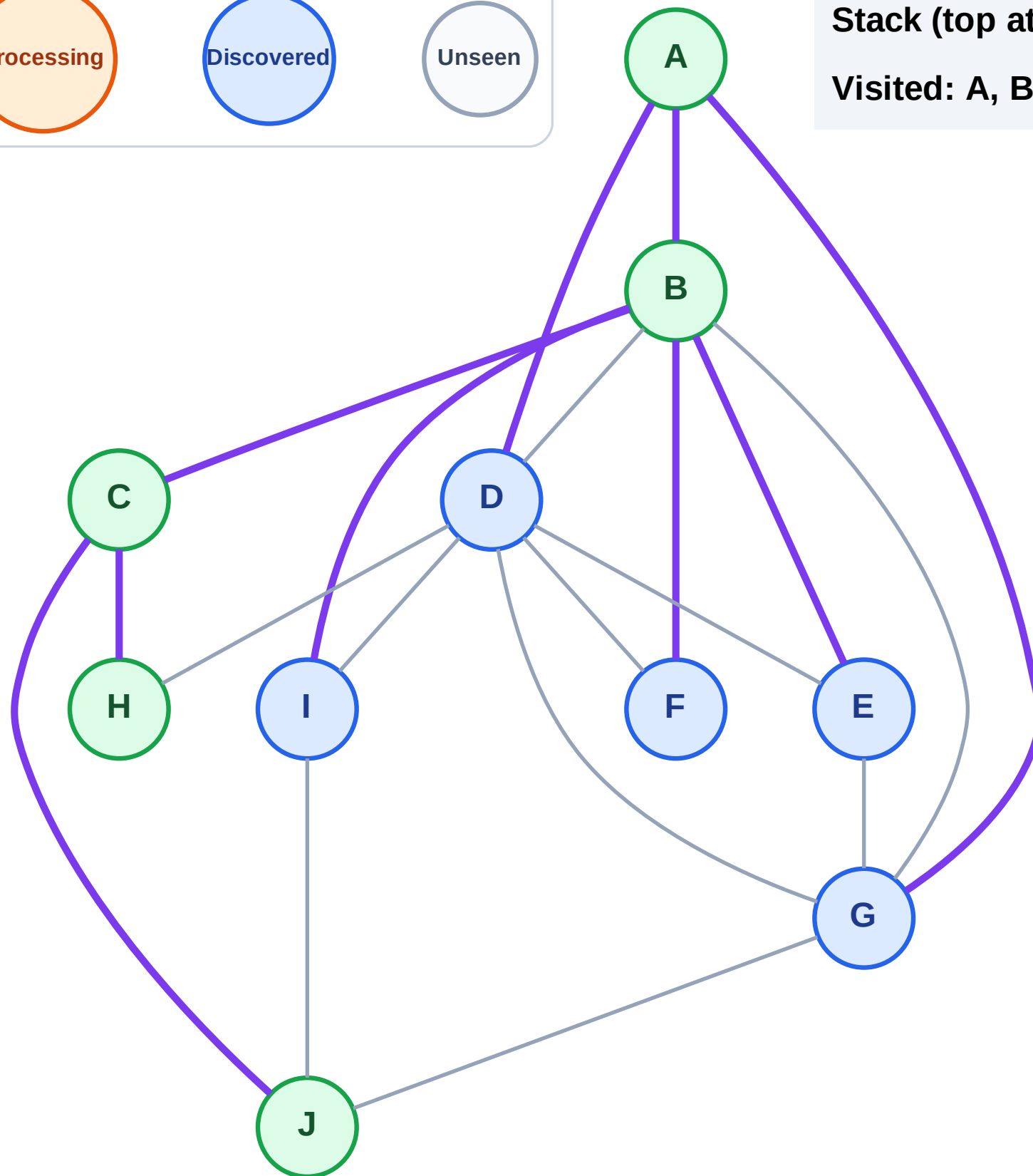
Step 11: No new neighbors from J

Legend



Stack (top at right): G | D | I | F | E

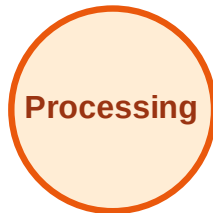
Visited: A, B, C, H, J



DFS Traversal

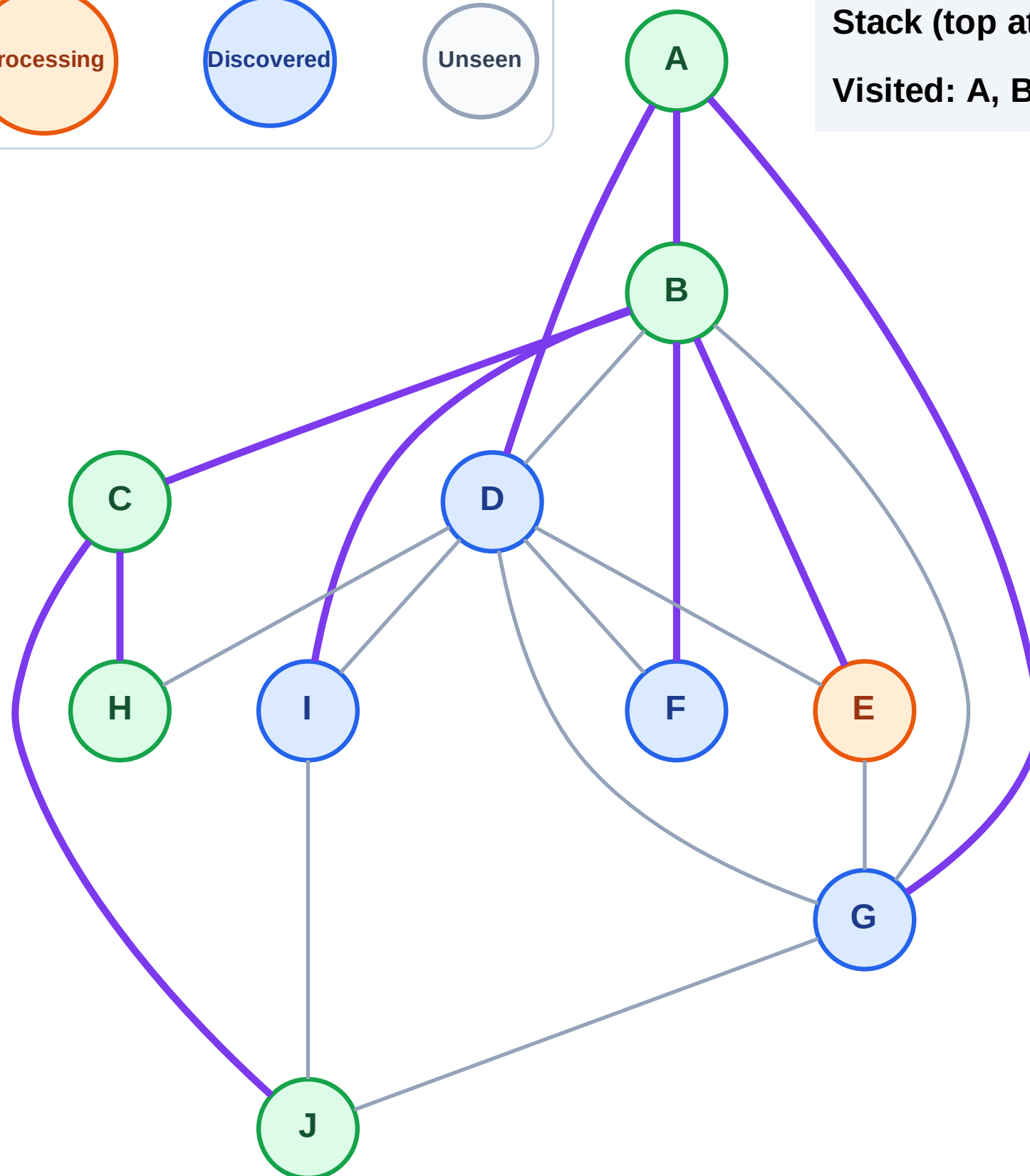
Step 12: Process node E

Legend



Stack (top at right): G | D | I | F

Visited: A, B, C, H, J



DFS Traversal

Step 13: No new neighbors from E

Legend

Complete

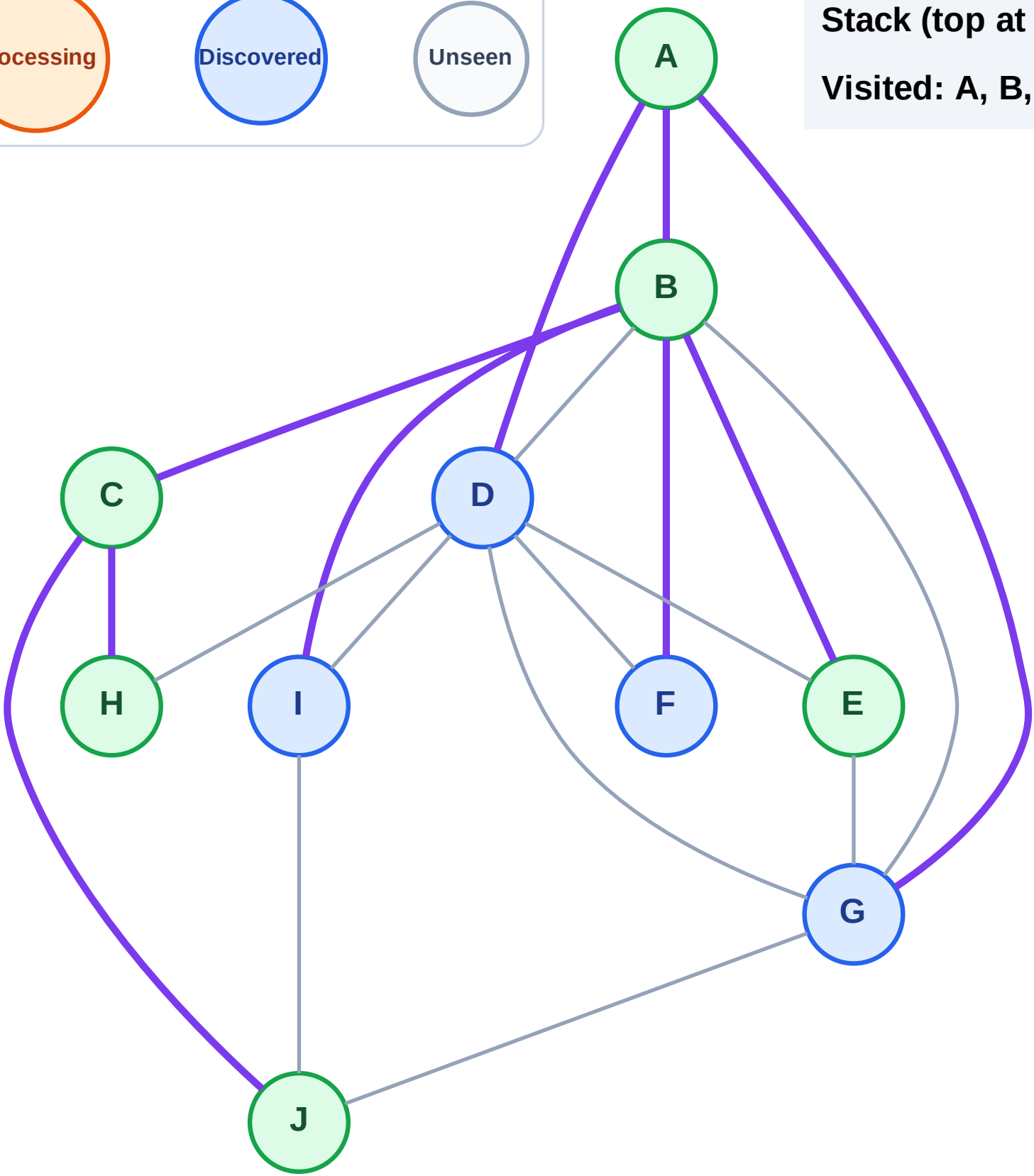
Processing

Discovered

Unseen

Stack (top at right): G | D | I | F

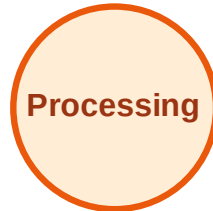
Visited: A, B, C, E, H, J



DFS Traversal

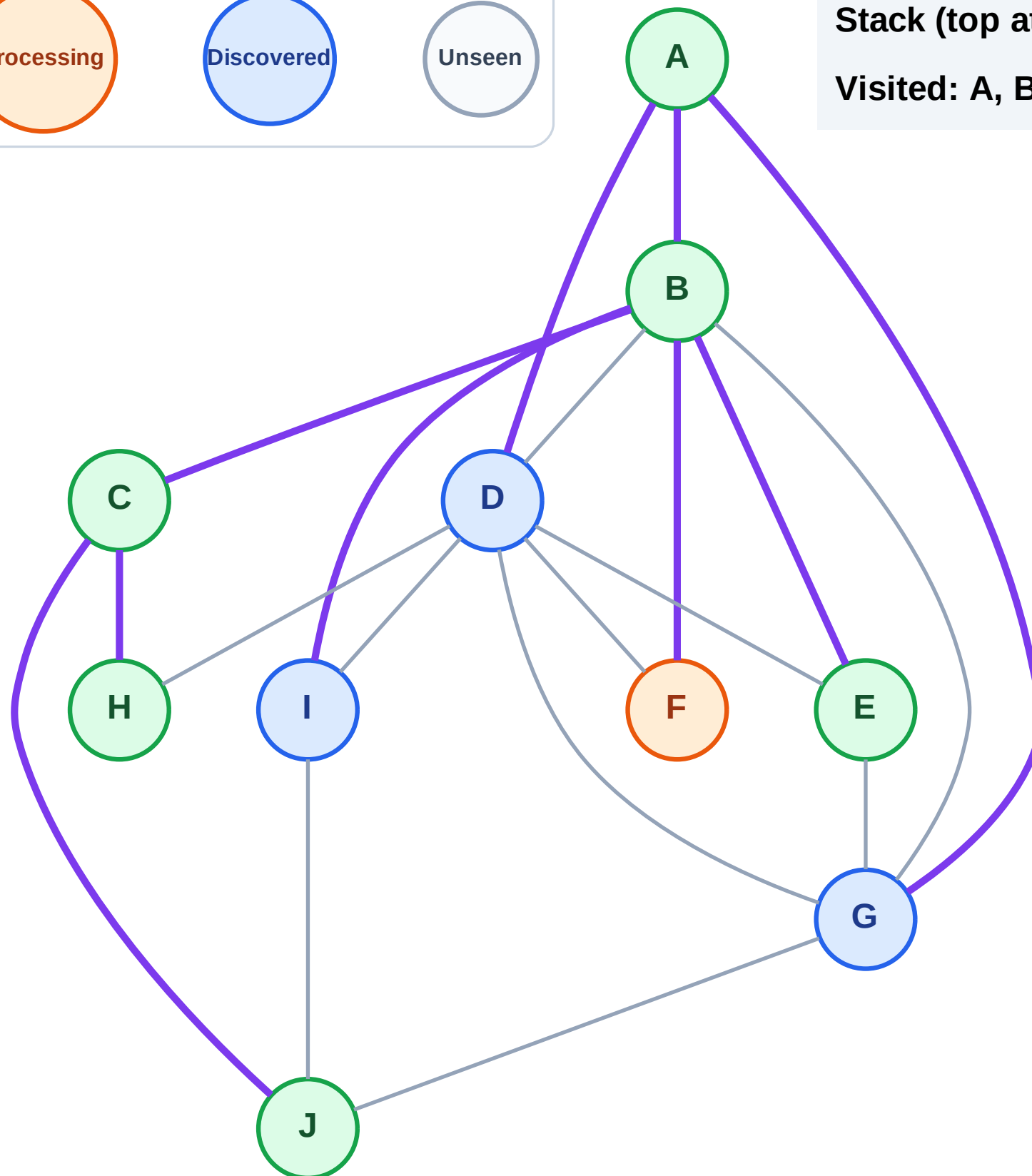
Step 14: Process node F

Legend



Stack (top at right): G | D | I

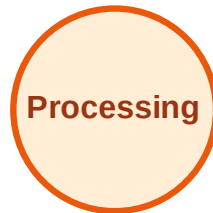
Visited: A, B, C, E, H, J



DFS Traversal

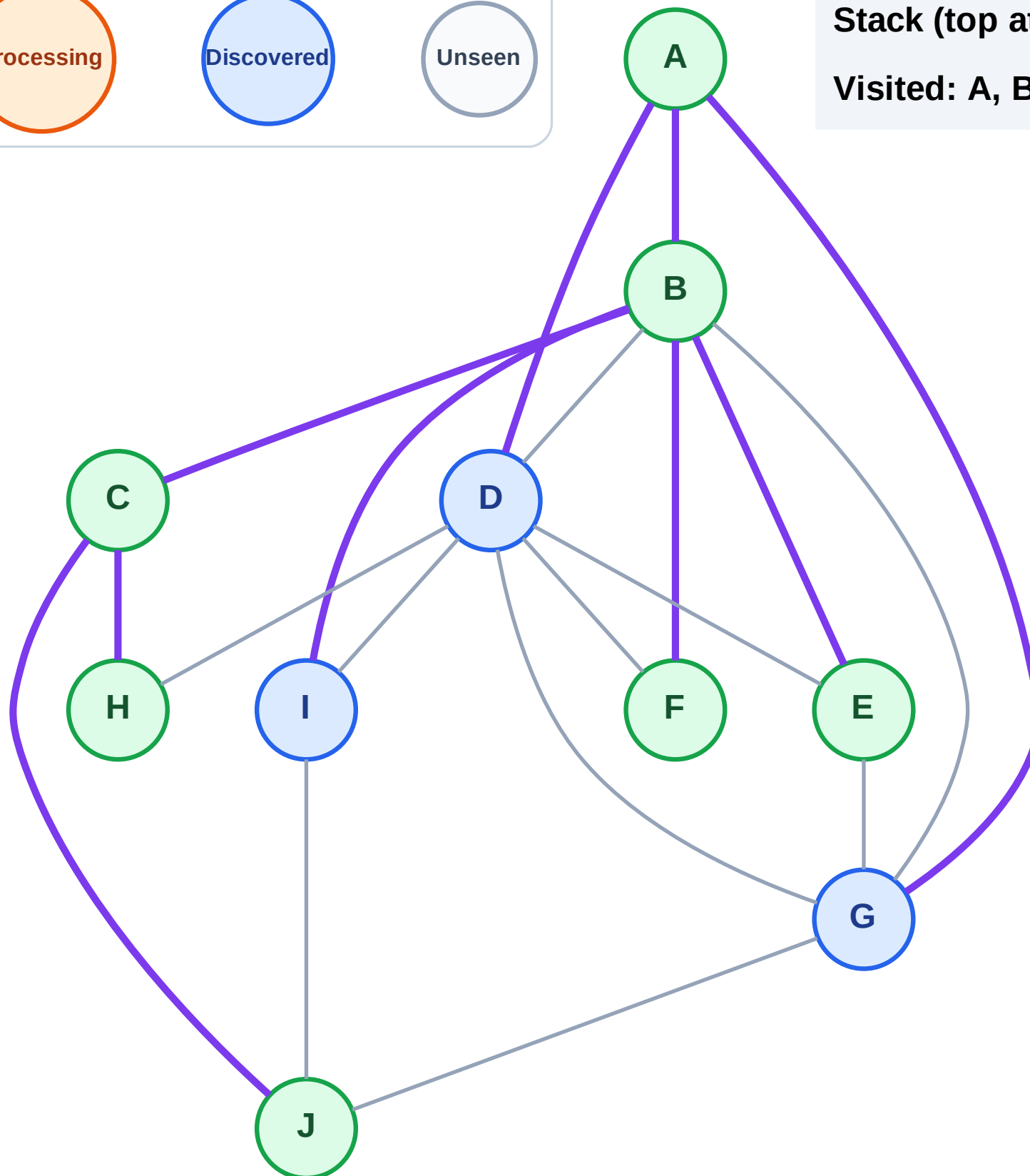
Step 15: No new neighbors from F

Legend



Stack (top at right): G | D | I

Visited: A, B, C, E, F, H, J



DFS Traversal

Step 16: Process node I

Legend

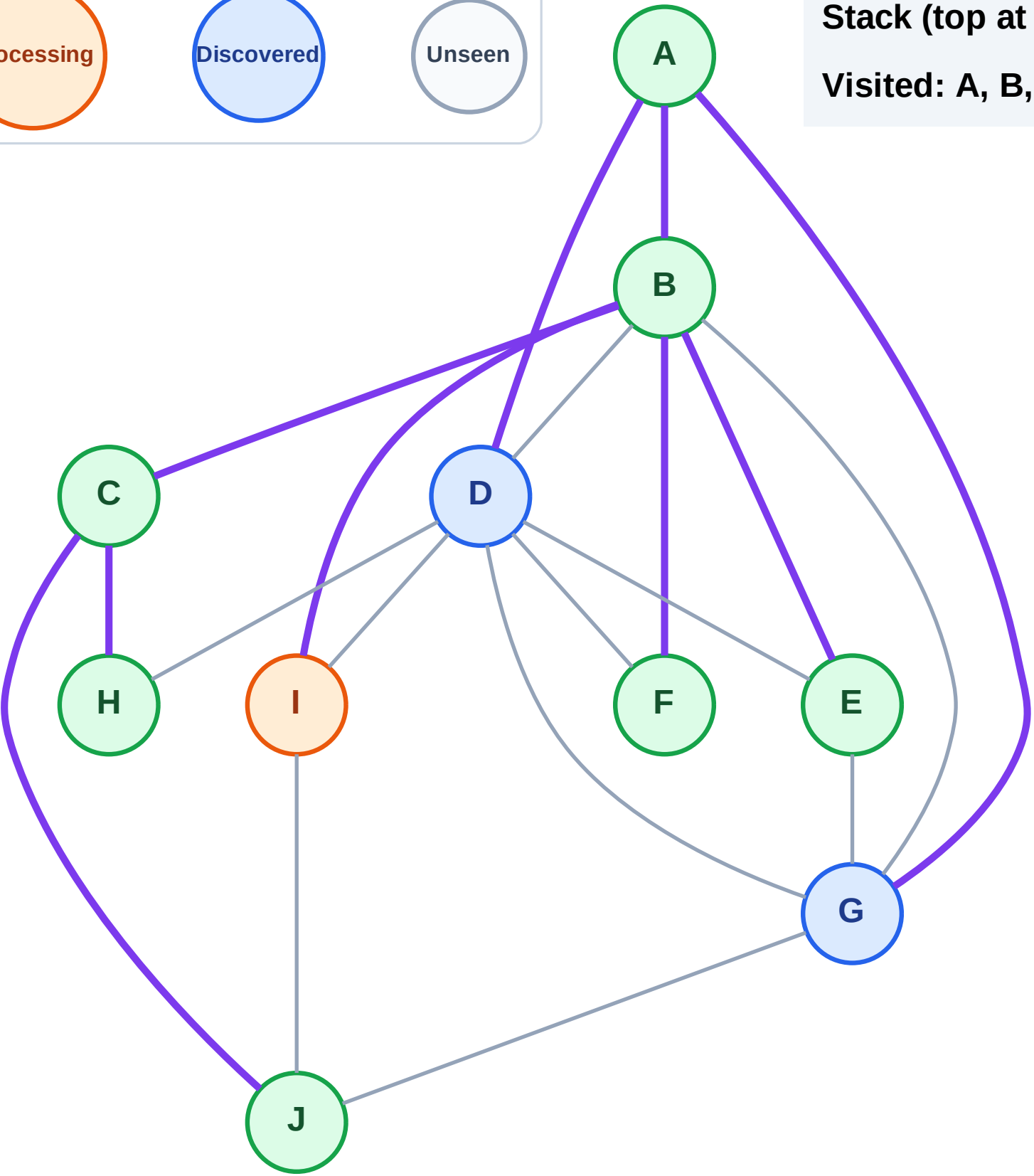
Complete

Processing

Discovered

Unseen

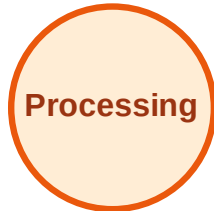
Stack (top at right): G | D
Visited: A, B, C, E, F, H, J



DFS Traversal

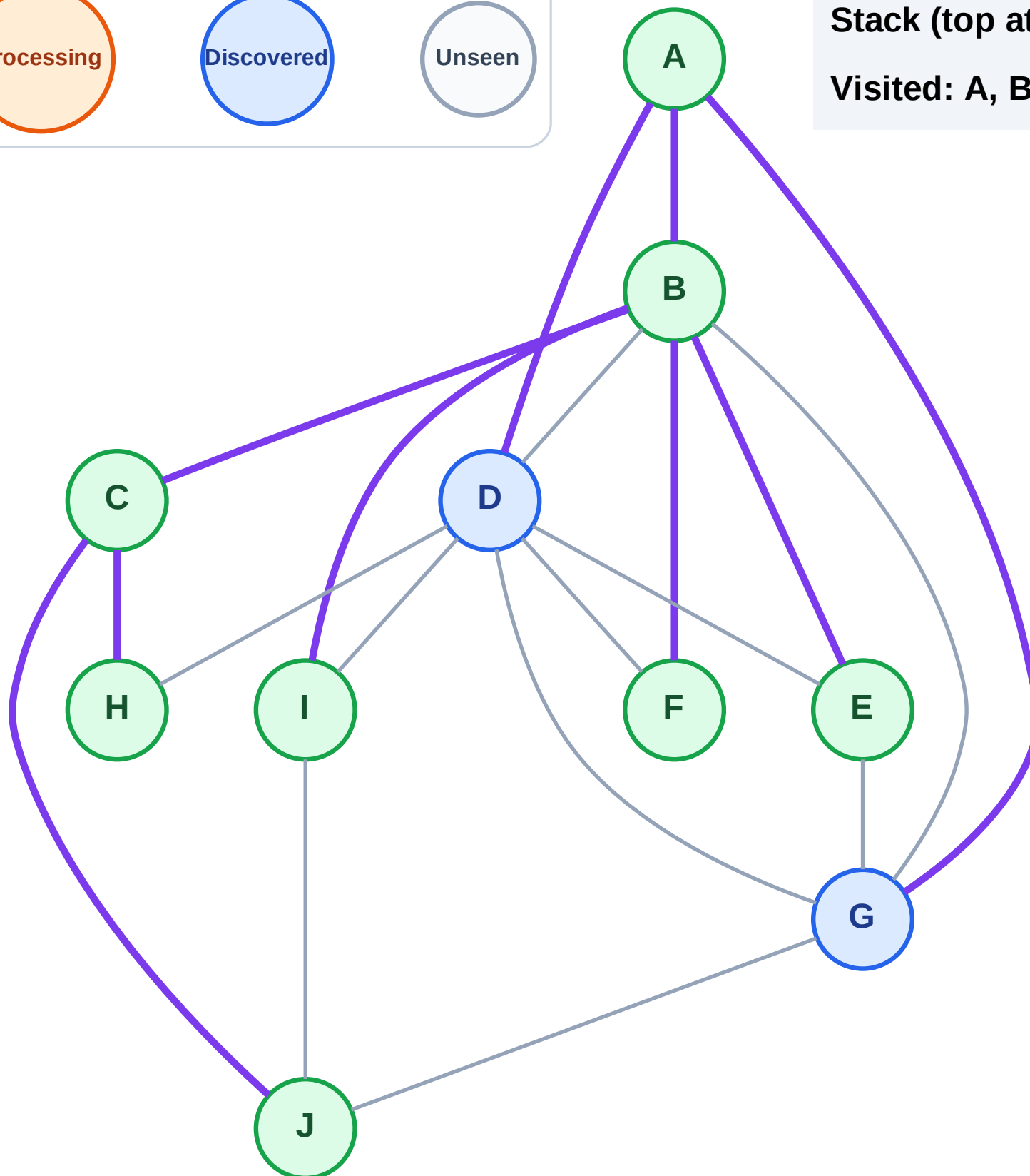
Step 17: No new neighbors from I

Legend



Stack (top at right): G | D

Visited: A, B, C, E, F, H, I, J



DFS Traversal

Step 18: Process node D

Legend

Complete

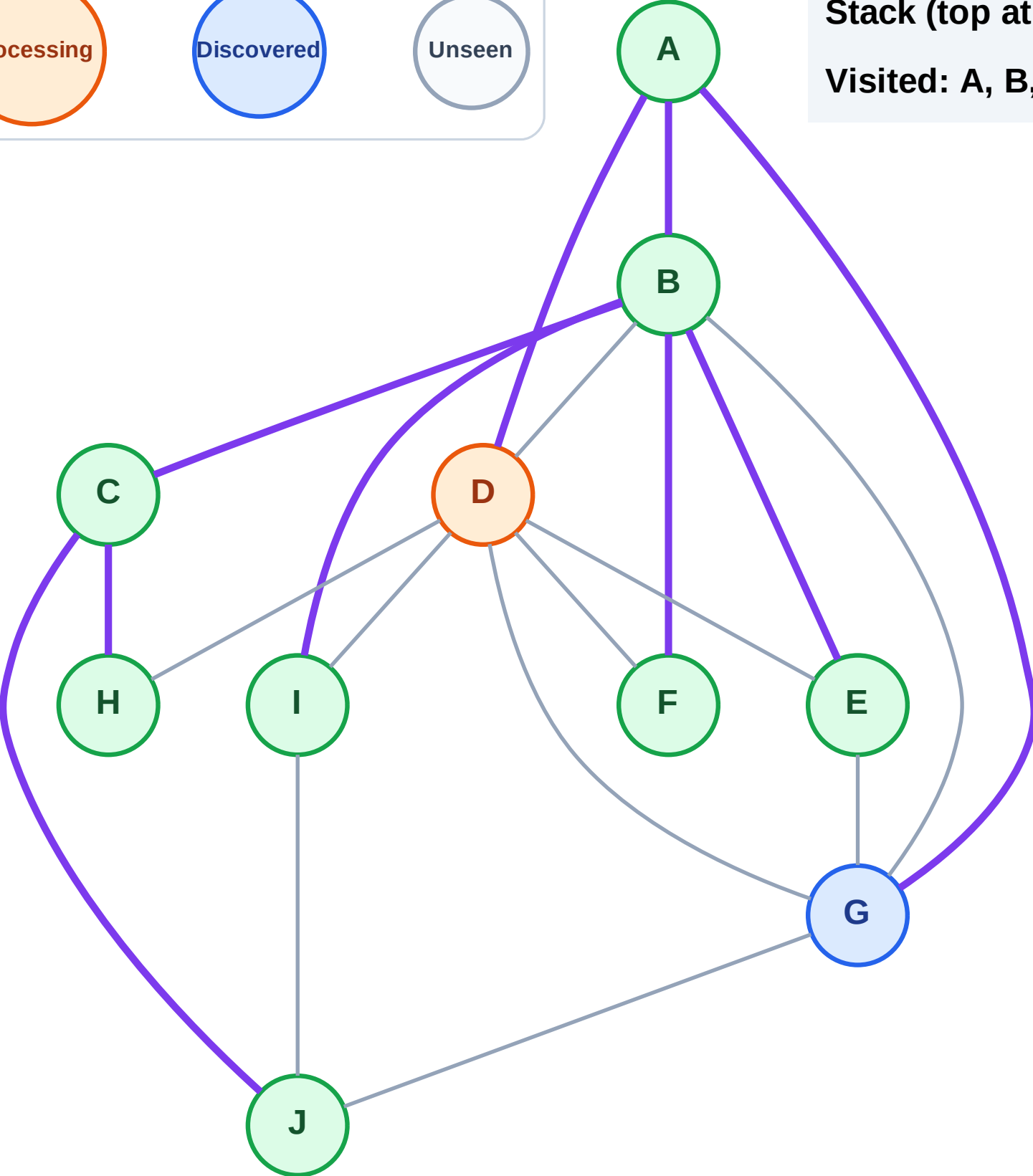
Processing

Discovered

Unseen

Stack (top at right): G

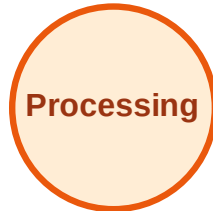
Visited: A, B, C, E, F, H, I, J



DFS Traversal

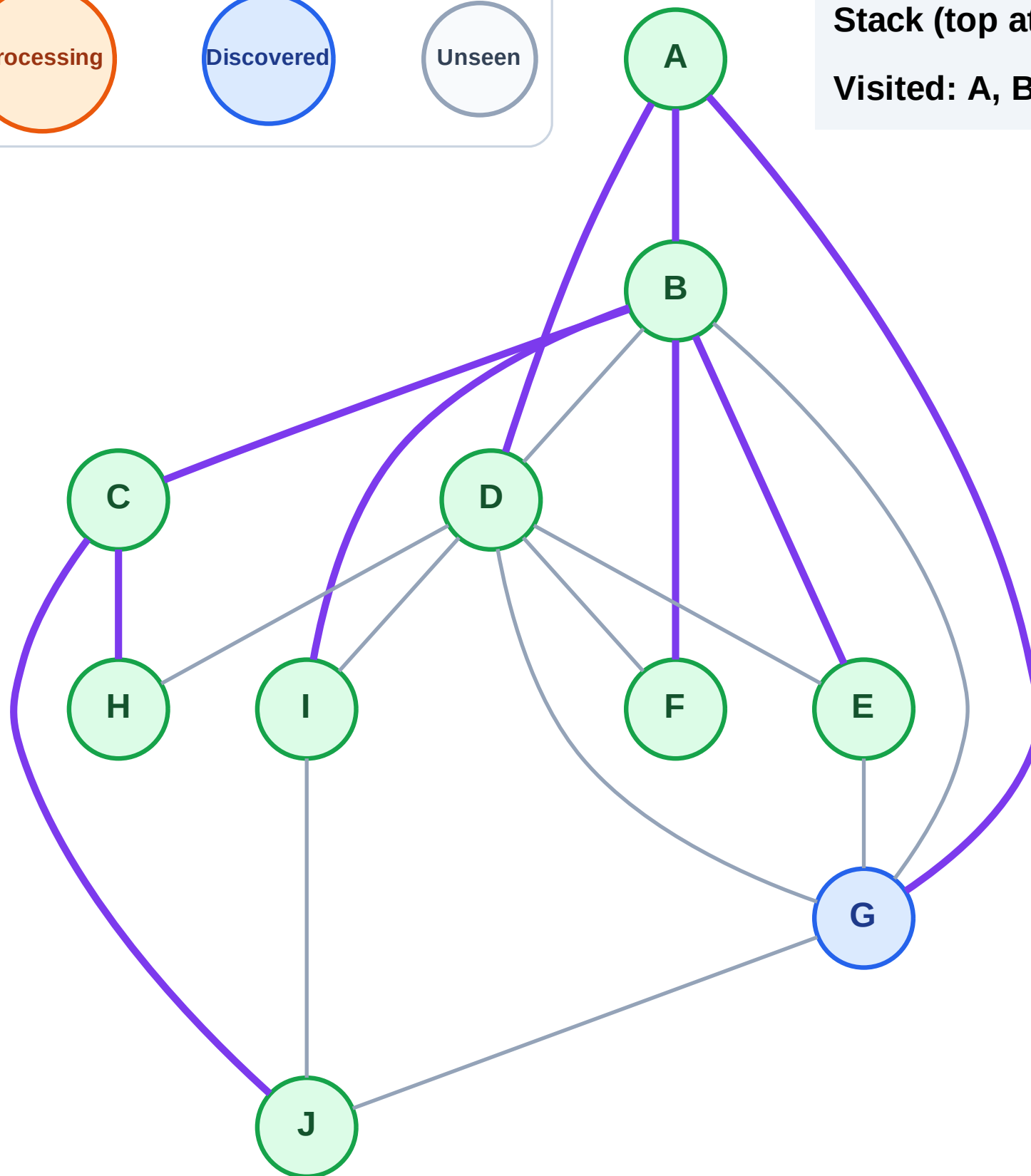
Step 19: No new neighbors from D

Legend



Stack (top at right): G

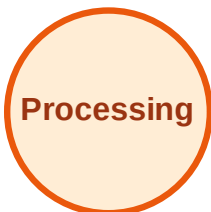
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

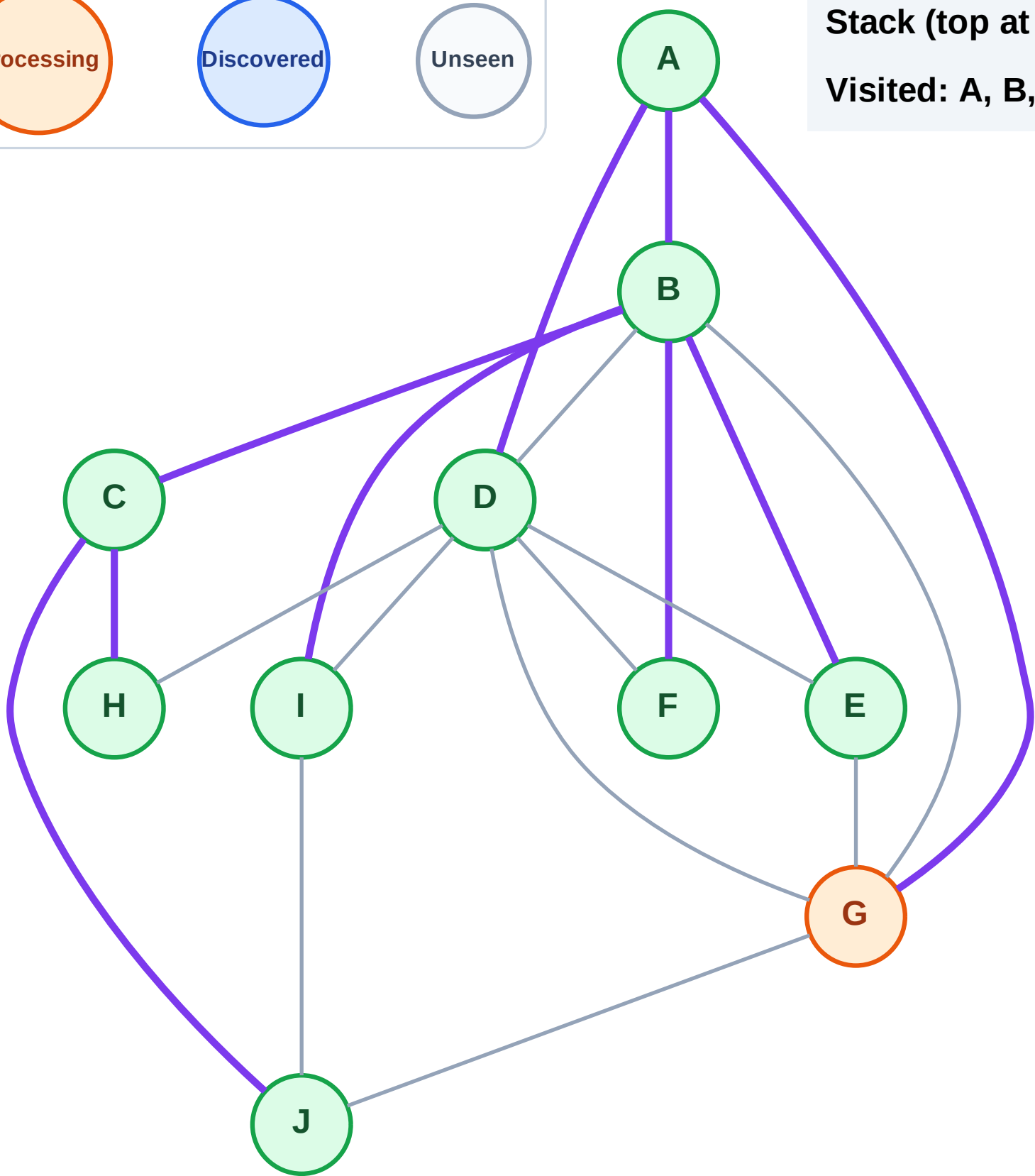
Step 20: Process node G

Legend



Stack (top at right): (empty)

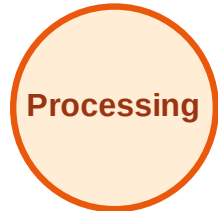
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

